

Opinion Perception Survey Analysis (preregistration)

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Invalid Date

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```
load_or_install <- function(pkg) {
  if (!require(pkg, character.only = TRUE)) {
    install.packages(pkg)
    library(pkg, character.only = TRUE)
  }
}

pkgs <- c(
  "tidyverse",
  "lme4",
  "lmerTest",
  "nloptr",
  "httpgd",
  "languageserver",
  "lmttest",
  "sandwich",
  "caret",
```

```

"knitr",
"prettycode",
"performance",
"estimatr",
"patchwork"
)
knitr::opts_chunk$set(comment = NA) # removes the ## prefix from output
invisible(lapply(pkgs, load_or_install))
options(vsc.viewer = TRUE)

```

1 Hypotheses and analysis plan

All hypotheses concerning a single slope parameter (e.g., H1 to H3) are tested using a t-test at significance level $\alpha = 0.05$. Hypotheses involving multiple parameters are tested using a single F-test on their joint significance, also at $\alpha = 0.05$. All subsequent hypotheses follow this convention without further repetition.

```

questions_sc = c(
  "climate_concern",
  "gay_marriage",
  "rights_indep_integration",
  "econ_inequality",
  "regulate_internet",
  "east_germans"
)
parties = c("Left Party", "BSW", "Green Party", "SPD", "FDP", "CDU/CSU", "AfD", "No party", "Other party", "Refuse to answer")
partiesVars = c("LeftParty", "BSW", "GreenParty", "SPD", "FDP", "CDU/CSU", "AfD")
partiesCloseSel = c("Left Party", "BSW", "Green Party", "SPD", "FDP", "CDU/CSU", "AfD")

```

2 Load Data

```

df_p <- read.csv("processed_data/2026-06-19_data_processed_participant_withAllIssueWeights.csv")

inds_bothwaves <- df_p %>%
  count(id) %>%
  filter(n == 2) %>%
  pull(id)

# FOR LATER
treatment_ids <- df_p |>
  filter(wave==2, treatment_wave2==TRUE) |>
  filter(id %in% inds_bothwaves) |>
  pull(id)
control_ids <- df_p |>
  filter(wave==2, treatment_wave2==FALSE) |>
  filter(id %in% inds_bothwaves) |>
  pull(id)

# Overall social circle std
df_p <- df_p %>%
  mutate(overall_std_socialCircle = rowMeans(
    across(all_of(paste0("std_socialCircle_ops_", questions_sc))),
  )) %>%
  mutate(across(
    all_of(paste0("std_socialCircle_ops_", questions_sc)),
    ~ .x^2,
    .names = "var_{gsub('std_', '', {.col})}"
  )) %>%
  mutate(wave = factor(wave))

var_cols <- paste0("var_socialCircle_ops_", questions_sc)
std_cols <- paste0("std_socialCircle_ops_", questions_sc)

df_diff <- read.csv("processed_data/2026-06-19_data_processed_differences_withAllIssueWeights.csv") %>%

```

```
mutate(ingroupdummy = as.integer(ingroupdummy == "True"))%>%
mutate(wave = factor(wave)) %>%
  left_join(
    df_p %>% select(id, wave, all_of(var_cols), all_of(std_cols)),
    by = c("id", "wave")
  )
# View(df_diff)
```

3 Hypotheses

3.1 Aim 1

Aim 1: Compare approaches to measuring perceived political differences

3.1.1 H1

3.1.1.1 H1. Participants place the typical voter of the party they affiliate with nearer to themselves than other typical voters.

We model the distance between the dots representing the participant (self) and the typical voter of each party r as

$$d_{n,\text{self},r} = \beta_0 + \beta_1 \mathbb{I}(r = \text{party}_n) + u_n + \epsilon_{n,r}, \tag{1}$$

where $\mathbb{I}(\cdot)$ is an indicator function, u_n is a participant level random intercept, and $\epsilon_{n,r}$ is an error term.

The prediction is $\beta_1 < 0$. Participants who prefer a political party place the typical voter of that party closer to themselves than the average of the typical voters of other parties.

```
df_h1 <- df_diff %>%
  filter(wave %in% c(1,2)) %>%
  filter(dot1 == "self", dot2 %in% partiesVars) %>%
  select(id, wave, ingroupdummy, party, pixel_dist)

model <- lmer(pixel_dist ~ ingroupdummy + (1 | id) + wave, data = df_h1, REML = TRUE)
intercept <- fixef(model)["(Intercept)"]
beta1 <- fixef(model)["ingroupdummy"]
group_var <- as.data.frame(VarCorr(model))$vcov[1]

# t-test on
coef_summary <- summary(model)$coefficients
t_stat <- coef_summary["ingroupdummy", "t value"]

# Satterthwaite degrees of freedom for p-value
df_satterthwaite <- coef_summary["ingroupdummy", "df"]
p_val <- coef_summary["ingroupdummy", "Pr(>|t|)"]
p_one_sided <- ifelse(beta1 < 0, p_val / 2, 1 - p_val / 2)
ci <- confint(model, parm = "ingroupdummy", method = "Wald")

cat("=== H1 Test (one-sided t-test on ) ===\n")
cat(paste0(
  sprintf(
    "d = %.2f + (%.2f) I(r=party_n) + u_n (with Var %.2f) + eps\n",
    intercept, beta1, group_var
  )))
cat(sprintf(" 1 = %.4f \n", beta1 ))
cat(sprintf(" t = %.4f\n", t_stat))
cat(sprintf(" df = %.2f\n", df_satterthwaite))
cat(sprintf(" p = %.4f (one-sided)\n", p_one_sided))
cat(sprintf(" 95% CI: [%.4f, %.4f]\n", ci[1], ci[2]))
cat(sprintf(" Reject H0 ( 1 < 0): %s\n",
  ifelse(p_one_sided < 0.05 & beta1 < 0, "YES ", "NO ")))
```

```
=== H1 Test (one-sided t-test on ) ===
d = 0.38 + (-0.25) I(r=party_n) + u_n (with Var 0.01) + eps
1 = -0.2520
t = -61.5081
df = 13785.65
```

```
p      = 0.0000 (one-sided)
95% CI: [-0.2600, -0.2440]
Reject H0 ( 1 < 0): YES
```

3.1.2 H2

3.1.2.1 H2. Distances to voters on the map correspond to likeability ratings.

We model likeability ratings of the typical voter of political party r (rated only in wave 2) as a function of map distances:

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + u_n + \epsilon_{n,r}. \quad (2)$$

The prediction is $\beta_1 < 0$. Voters placed farther away from the participant's own dot on the map receive lower likeability ratings than voters placed closer to the participant's own dot.

```
df_h2 <- df_diff %>%
  filter(dot1 == "self", dot2 %in% partiesVars, wave == 2) %>%
  select(id, wave, sympathy, party, pixel_dist)

model <- lmer(sympathy ~ pixel_dist + (1 | id), data = df_h2, REML = TRUE)

intercept <- fixef(model)["(Intercept)"]
beta1     <- fixef(model)["pixel_dist"]
group_var <- as.data.frame(VarCorr(model))$vcov[1]

coef_summary <- summary(model)$coefficients
t_stat       <- coef_summary["pixel_dist", "t value"]
p_val        <- coef_summary["pixel_dist", "Pr(>|t|)"]

p_one_sided <- ifelse(beta1 < 0, p_val / 2, 1 - p_val / 2)
ci          <- confint(model, parm = "pixel_dist", method = "Wald")

cat("=== H2 Test (one-sided t-test on  ) ===\n")
cat(sprintf("Model: y_r = %.2f + (%.2f) d_self-r + u_n (Var=%.2f) + eps\n\n", intercept, beta1, group_var))
cat(sprintf("      = %.4f\n",          beta1))
cat(sprintf("    t      = %.4f\n",          t_stat))
cat(sprintf("    p      = %.4f (one-sided)\n",  p_one_sided))
cat(sprintf(" 95% CI: [%.4f, %.4f]\n",      ci[1], ci[2]))
cat(sprintf(" Reject H0 (  < 0): %s\n",    ifelse(p_one_sided < 0.05 & beta1 < 0, "YES ", "NO  "))
```

```
=== H2 Test (one-sided t-test on  ) ===
Model: y_r = 0.79 + (-1.11) d_self-r + u_n (Var=0.02) + eps

      = -1.1108
    t      = -69.6313
    p      = 0.0000 (one-sided)
 95% CI: [-1.1421, -1.0795]
Reject H0 (  < 0): YES
```

3.1.3 H3

3.1.3.1 H3. Distances on the political map correspond to similarity ratings.

Similar to H2, we model similarity ratings between a pair of individuals, i and j , as a function of map distances:

$$s_{n,ij} = \beta_0 + \beta_1 \cdot d_{n,ij} + u_n + \epsilon_{n,ij}. \quad (3)$$

The prediction is $\beta_1 < 0$. Pairs placed farther apart on the map receive lower similarity ratings than pairs placed closer together.

```
df_h3 <- df_diff %>%
  filter(wave %in% c(1,2)) %>%
  select(id, wave, pairwise_similarity, pixel_dist) %>%
  drop_na()

model <- lmer(pairwise_similarity ~ pixel_dist + (1 | id) + wave , data = df_h3, REML = TRUE)

intercept <- fixef(model)["(Intercept)"]
beta1     <- fixef(model)["pixel_dist"]
```

```
group_var <- as.data.frame(VarCorr(model))$vcov[1]

coef_summary <- summary(model)$coefficients
t_stat <- coef_summary["pixel_dist", "t value"]
df_satterthwaite <- coef_summary["pixel_dist", "df"]
p_val <- coef_summary["pixel_dist", "Pr(>|t|)"]

p_one_sided <- ifelse(beta1 < 0, p_val / 2, 1 - p_val / 2)
ci <- confint(model, parm = "pixel_dist", method = "Wald")

cat("=== H3 Test (one-sided t-test on ) ===\n")
cat(paste0(sprintf("Model: s_ij = %.2f + (%.2f) d_ij + u_n (Var = %.2f) + eps\n", intercept, beta1, group_var)))
cat(sprintf("      = %.4f\n", beta1))
cat(sprintf(" t      = %.4f\n", t_stat))
cat(sprintf(" df      = %.2f\n", df_satterthwaite))
cat(sprintf(" p      = %.4f (one-sided)\n", p_one_sided))
cat(sprintf(" 95% CI: [%.4f, %.4f]\n", ci[1], ci[2]))
cat(sprintf(" Reject H0 ( < 0): %s\n", ifelse(p_one_sided < 0.05 & beta1 < 0, "YES ", "NO ")))
```

```
=== H3 Test (one-sided t-test on ) ===
Model: s_ij = 0.66 + (-0.78) d_ij + u_n (Var = 0.01) + eps
      = -0.7761
 t      = -117.4882
 df      = 38233.23
 p      = 0.0000 (one-sided)
 95% CI: [-0.7890, -0.7631]
 Reject H0 ( < 0): YES
```

3.1.4 H4

3.1.4.1 H4. The closest pair on the map is rated most similar and the farthest pair is rated least similar

Consider $(ij)^{min}$ the pair of individuals i and j with the smallest distance on participant n 's political map and $(ij)^{max}$ the pair with the largest distance on the map. The similarity difference between these two pairs

$$\Delta s_{max} = s_{n,(ij)^{min}} - s_{n,(ij)^{max}} . \quad (4)$$

The prediction is $\Delta s_{max} > 0$ (standard t-test with $\alpha = 0.05$). The closest pair on the map receives a higher similarity rating than the farthest pair.

```
df_h4 <- df_diff %>%
  filter(wave %in% c(1,2)) %>%
  filter(minPixelDistance_pair_dummy == "True" | maxPixelDistance_pair_dummy == "True") %>%
  select(id, wave, minPixelDistance_pair_dummy, maxPixelDistance_pair_dummy, pairwise_similarity)

df_h4_wide <- df_h4 %>%
  pivot_wider(
    id_cols = c(id, wave),
    names_from = maxPixelDistance_pair_dummy,
    values_from = pairwise_similarity
  ) %>%
  rename(sim_min = "False", sim_max = "True") %>%
  mutate(diff = sim_min - sim_max)

t_result <- t.test(df_h4_wide$diff, mu = 0)
t_stat <- t_result$statistic
p_two_tailed <- t_result$p.value
p_one_tailed <- p_two_tailed / 2

cat("=== H4 Test (one-sample t-test on smax): ===\n")
cat(sprintf(" Mean diff (s_minDist - s_maxDist) = %.4f (+-.%.4f) - %.4f (+-.%.4f) %.4f\n", mean(df_h4_wide$sim_min),
cat(sprintf(" t      = %.4f\n", t_stat))
cat(sprintf(" p (two-tailed) = %.4f\n", p_two_tailed))
cat(sprintf(" p (one-tailed) = %.4f\n", p_one_tailed))
cat(sprintf(" H4 supported: %s\n", t_stat > 0 && p_one_tailed < 0.05)))
```

```
=== H4 Test (one-sample t-test on smax): ===
Mean diff (s_minDist - s_maxDist) = 0.7113 (+-0.2688) - 0.2308 (+-0.2771) 0.4805
t      = 55.3777
p (two-tailed) = 0.0000
p (one-tailed) = 0.0000
H4 supported: TRUE
```

3.1.5 H5

3.1.5.1 H5. The inferred issue weights are robust across waves.

We test the reliability of our tool by comparing the similarity of issue weights, $\alpha_{n,q}$, of participants in wave 1 vs.~wave 2 (control condition) with the similarity of issue weights between participants.

$$\overline{\Delta\alpha}_{q_{\text{within}}} = \mathbb{E} [|\alpha_{n,q}(\text{wave 2}) - \alpha_{n,q}(\text{wave 1})|] \quad \text{and} \tag{5}$$

$$\overline{\Delta\alpha}_{q_{\text{between}}} = \mathbb{E} [|\alpha_{n,q}(\text{wave 2}) - \alpha_{m,q}(\text{wave 1/wave 2})|] \quad \text{with } m \neq n \tag{6}$$

ERROR: It should be: $\overline{\Delta\alpha}_{q_{\text{within}}} - \overline{\Delta\alpha}_{q_{\text{between}}} < 0$

The prediction is that $\overline{\Delta\alpha}_{q_{\text{within}}} - \overline{\Delta\alpha}_{q_{\text{between}}} > 0$ (one-sample t-test with $\alpha = 0.05$). Participants' issue weights in wave 2 are more similar to their own issue weights in the previous wave than to those of other participants in wave 1 or wave 2 (in the control condition).

```
k <- "exp"
alpha_cols <- paste0(k, "_alpha_", questions_sc)

control_ids <- df_p %>%
  filter(id %in% inds_bothwaves, wave == 2, treatment_wave2 == "0") %>%
  pull(id)

df_h5 <- df_p %>%
  filter(if_all(all_of(alpha_cols), ~ !is.na(.x))) %>%
  filter(id %in% control_ids) %>%
  select(id, wave, all_of(alpha_cols))

# Keep only ids present in BOTH waves
ids_both <- intersect(
  df_h5 %>% filter(wave == 1) %>% pull(id),
  df_h5 %>% filter(wave == 2) %>% pull(id)
)
df_h5 <- df_h5 %>% filter(id %in% ids_both)

w1 <- df_h5 %>% filter(wave == 1) %>% arrange(id) %>% select(id, all_of(alpha_cols))
w2 <- df_h5 %>% filter(wave == 2) %>% arrange(id) %>% select(id, all_of(alpha_cols))

w1_mat <- w1 %>% select(all_of(alpha_cols)) %>% as.matrix()
w2_mat <- w2 %>% select(all_of(alpha_cols)) %>% as.matrix()
ids     <- w1 %>% pull(id)
n       <- length(ids)

# Within: each participant's w2 vs their own w1
within_means <- rowMeans(abs(w2_mat - w1_mat))

# Between: each participant's w2 vs all other participants' w1
between_means <- numeric(n)
for (i in seq_len(n)) {
  other <- w2_mat[-i, , drop = FALSE]
  diffs <- rowMeans(abs(sweep(other, 2, w1_mat[i, ])))
  between_means[i] <- mean(diffs)
}

delta <- within_means - between_means
t_result <- t.test(delta, mu = 0, alternative = "two.sided")
t_stat <- t_result$statistic
p_two_tailed <- t_result$p.value
p_one_tailed <- p_two_tailed / 2
```

```
cohens_d      <- mean(delta) / sd(delta)
supported     <- t_stat < 0 && p_one_tailed < 0.05

cat("=== H5 Test ===\n")
cat(sprintf(" ...using alpha_%s and Manhattan distance\n", k))
cat(sprintf(" Within (mean ± sd):  %.4f ± %.4f\n", mean(within_means), sd(within_means)))
cat(sprintf(" Between (mean ± sd):  %.4f ± %.4f\n", mean(between_means), sd(between_means)))
cat(sprintf(" Delta (mean):          %.4f\n\n", mean(delta)))
cat(sprintf(" t = %.4f\n", t_stat))
cat(sprintf(" p (two) = %.4f\n", p_two_tailed))
cat(sprintf(" p (one) = %.4f\n", p_one_tailed))
cat(sprintf(" Cohen's d = %.4f\n\n", cohens_d))
cat(sprintf(" H5 supported: %s\n", ifelse(supported, "YES ", "NO ")))
```

```
=== H5 Test ===
...using alpha_exp and Manhattan distance
Within (mean ± sd):  0.1988 ± 0.0777
Between (mean ± sd):  0.2211 ± 0.0259
Delta (mean):        -0.0223

t          = -6.1494
p (two)    = 0.0000
p (one)    = 0.0000
Cohen's d  = -0.3094

H5 supported: YES
```

3.1.6 Exploration
3.1.6.1 Explorative Analyses.

In addition to the preregistered hypotheses above, we will explore how people rate the political mapping task against the pairwise similarity task in terms of difficulty and enjoyment, how satisfied they are with the political map, and how much time they needed for each task.
(see 03_data-overview)

3.2 Aim 2
Aim 2: Identify patterns linked to (political) identity

In this section, we hypothesise that demographic and identity-related variables, including party preference, party_n, left-right placement, lr_n, age, age_n, or combinations of them moderate the relationship between belief differences and issue weights.

3.2.1 H6
3.2.1.1 H6. Party preference predicts issue weights.

We model the (normalised) issue weights for each issue q as

alpha_{n,q} = 1/6 + sum_r beta_{r,q} * I[party_n = r] + u_q + epsilon_{n,q} (7)

where I[party_n = r] are dummy variables for each party r (with one party as reference category), beta_{r,q} is the effect of party r on weight assigned to issue q, u_q is the average deviation of the weight for issue q, and epsilon_{n,q} is the individual- and issue-specific error.

The prediction is beta_{r,q} != 0 and beta_{r1,q} != beta_{r2,q} for at least some parties r1 and r2. Participants affiliated with different parties differ in how strongly they weigh different political issues when mapping individuals.

3.2.1.2 Interpretation 1: Do parties differ across issues?

For correlations (unconstrained):

```
k      <- "corrS"  # ONLY FOR CORRELATIONS
alpha_cols <- paste0(k, "_alpha_", questions_sc)

# --- Data ---
df_h6 <- df_p %>%
  filter(wave %in% c(1, 2)) %>%
  select(id, wave, party_close, all_of(alpha_cols)) %>%
```

```

filter(party_close %in% c(partiesCloseSel, "No party")) %>%
pivot_longer(
  cols      = all_of(alpha_cols),
  names_to  = "issue",
  names_prefix = paste0(k, "_alpha_"),
  values_to = "alpha_q"
) %>%
filter(!is.na(party_close), !is.na(alpha_q)) %>%
mutate(
  party_close = relevel(factor(party_close), ref = "No party"),
  issue       = relevel(factor(issue),      ref = "climate_concern"),
  wave       = factor(wave) # treat as categorical; only 2 levels
)

# --- Models ---
# Random intercept per person (1|id) accounts for repeated measures across waves/issues.
# Wave is a fixed effect covariate to control for time.
model_h6_null <- lmer(alpha_q ~ issue + wave + (1 | id),          data = df_h6)
model_h6_full <- lmer(alpha_q ~ party_close * issue + wave + (1 | id), data = df_h6)

# --- Model comparison ---
wt <- anova(model_h6_null, model_h6_full)

```

refitting model(s) with ML (instead of REML)

```

r2_null <- r2(model_h6_null)
r2_full <- r2(model_h6_full)
df_null <- length(fixef(model_h6_null))
df_full <- length(fixef(model_h6_full))

ic_compare <- data.frame(
  Model      = c("Null (issue + wave)", "Full (party × issue + wave)"),
  AIC        = round(c(AIC(model_h6_null), AIC(model_h6_full)), 1),
  BIC        = round(c(BIC(model_h6_null), BIC(model_h6_full)), 1),
  logLik     = round(c(as.numeric(logLik(model_h6_null)), as.numeric(logLik(model_h6_full))), 1),
  delta_AIC  = round(c(AIC(model_h6_null), AIC(model_h6_full)) - min(c(AIC(model_h6_null), AIC(model_h6_full))), 1),
  delta_BIC  = round(c(BIC(model_h6_null), BIC(model_h6_full)) - min(c(BIC(model_h6_null), BIC(model_h6_full))), 1),
  npar       = c(df_null, df_full)
)

# --- Output ---
cat("=== H6: Party preference predicts issue weights ===\n")
cat("  (wave controlled for as fixed effect; random intercept per id)\n")

cat("Model fit: \n", sep = "")
print(ic_compare, row.names = FALSE)

cat(sprintf(
  "\nR2 marginal / conditional\n Null : %.3f / %.3f\n Full : %.3f / %.3f\n",
  r2_null$R2_marginal, r2_null$R2_conditional,
  r2_full$R2_marginal, r2_full$R2_conditional
))

cat("\nLikelihood ratio test (null vs. full):\n", sep = "")
print(wt)

cat("\nFixed effects - full model:\n", sep = "")
printCoefmat(summary(model_h6_full)$coefficients, digits = 3, signif.stars = TRUE)

cat("\nRandom effects - full model:\n", sep = "")
print(VarCorr(model_h6_full))

cat("Verdict - H6\n")
cat(sprintf(
  "  LRT party × issue interaction: 2(%d) = %.2f, p = %.3f\n",

```

```
wt$Df[2], wt$Chisq[2], wt$`Pr(>Chisq)`[2]
))
if (wt$`Pr(>Chisq)`[2] < .05) {
  cat(" → Supported: party affiliation significantly predicts issue weights\n")
} else {
  cat(" → Not supported: no significant effect of party on issue weights\n")
}
```

=== H6: Party preference predicts issue weights ===
(wave controlled for as fixed effect; random intercept per id)
Model fit:

	Model	AIC	BIC	logLik	delta_AIC	delta_BIC	npar
Null	(issue + wave)	-3880.3	-3813.5	1949.2	48	0.0	7
Full	(party × issue + wave)	-3928.3	-3549.7	2015.1	0	263.8	49

R² marginal / conditional
Null : 0.032 / 0.440
Full : 0.071 / 0.452

Likelihood ratio test (null vs. full):

Data: df_h6

Models:

model_h6_null:	alpha_q ~ issue + wave + (1 id)						
model_h6_full:	alpha_q ~ party_close * issue + wave + (1 id)						
	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df Pr(>Chisq)
model_h6_null	9	-3942.5	-3875.7	1980.2	-3960.5		
model_h6_full	51	-4255.5	-3877.0	2178.8	-4357.5	397.03	42 < 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Fixed effects - full model:

	Estimate	Std. Error
(Intercept)	2.77e-01	1.32e-02
party_closeAfd	5.59e-02	1.81e-02
party_closeBSW	2.55e-02	3.02e-02
party_closeCDU/CSU	-5.27e-02	1.73e-02
party_closeFDP	-2.08e-02	3.03e-02
party_closeGreen Party	1.30e-01	2.04e-02
party_closeLeft Party	1.33e-01	2.31e-02
party_closeSPD	5.69e-03	1.89e-02
issueeast_germans	-6.47e-02	1.43e-02
issueecon_inequality	-4.67e-02	1.43e-02
issuegay_marriage	-3.27e-02	1.42e-02
issueregulate_internet	-7.72e-02	1.42e-02
issuerights_indep_integration	2.73e-02	1.43e-02
wave2	5.83e-03	3.56e-03
party_closeAfd:issueeast_germans	-2.75e-02	1.98e-02
party_closeBSW:issueeast_germans	-1.67e-02	3.33e-02
party_closeCDU/CSU:issueeast_germans	4.45e-02	1.89e-02
party_closeFDP:issueeast_germans	3.86e-02	3.37e-02
party_closeGreen Party:issueeast_germans	-1.01e-01	2.10e-02
party_closeLeft Party:issueeast_germans	-1.72e-01	2.45e-02
party_closeSPD:issueeast_germans	-3.46e-02	2.02e-02
party_closeAfd:issueecon_inequality	-2.99e-02	1.98e-02
party_closeBSW:issueecon_inequality	-1.88e-02	3.33e-02
party_closeCDU/CSU:issueecon_inequality	1.75e-02	1.89e-02
party_closeFDP:issueecon_inequality	6.03e-03	3.36e-02
party_closeGreen Party:issueecon_inequality	-5.48e-02	2.10e-02
party_closeLeft Party:issueecon_inequality	-2.45e-02	2.45e-02
party_closeSPD:issueecon_inequality	7.61e-03	2.02e-02
party_closeAfd:issuegay_marriage	-6.16e-02	1.98e-02
party_closeBSW:issuegay_marriage	-2.47e-02	3.33e-02
party_closeCDU/CSU:issuegay_marriage	2.86e-02	1.89e-02
party_closeFDP:issuegay_marriage	2.58e-02	3.36e-02
party_closeGreen Party:issuegay_marriage	2.36e-02	2.10e-02

party_closeLeft Party:issuegay_marriage	1.44e-02	2.45e-02
party_closeSPD:issuegay_marriage	1.86e-02	2.02e-02
party_closeAfD:issueregulate_internet	-2.28e-02	1.98e-02
party_closeBSW:issueregulate_internet	-5.65e-02	3.33e-02
party_closeCDU/CSU:issueregulate_internet	1.62e-02	1.89e-02
party_closeFDP:issueregulate_internet	-4.71e-02	3.36e-02
party_closeGreen Party:issueregulate_internet	-1.00e-01	2.10e-02
party_closeLeft Party:issueregulate_internet	-1.47e-01	2.45e-02
party_closeSPD:issueregulate_internet	-1.86e-02	2.02e-02
party_closeAfD:issuerights_indep_integration	1.66e-02	1.98e-02
party_closeBSW:issuerights_indep_integration	-5.28e-02	3.33e-02
party_closeCDU/CSU:issuerights_indep_integration	1.03e-02	1.89e-02
party_closeFDP:issuerights_indep_integration	4.23e-02	3.36e-02
party_closeGreen Party:issuerights_indep_integration	-6.34e-02	2.10e-02
party_closeLeft Party:issuerights_indep_integration	-5.20e-02	2.45e-02
party_closeSPD:issuerights_indep_integration	-6.65e-02	2.02e-02

	df	t	value	Pr(> t)
(Intercept)	7.56e+03	20.92	< 2e-16	
party_closeAfD	8.63e+03	3.09	0.0020	
party_closeBSW	9.10e+03	0.84	0.3984	
party_closeCDU/CSU	8.53e+03	-3.05	0.0023	
party_closeFDP	9.48e+03	-0.69	0.4927	
party_closeGreen Party	5.10e+03	6.35	2.3e-10	
party_closeLeft Party	6.75e+03	5.76	8.9e-09	
party_closeSPD	7.25e+03	0.30	0.7639	
issueeast_germans	1.11e+04	-4.54	5.6e-06	
issueecon_inequality	1.11e+04	-3.28	0.0010	
issuegay_marriage	1.11e+04	-2.30	0.0216	
issueregulate_internet	1.11e+04	-5.42	5.9e-08	
issuerights_indep_integration	1.11e+04	1.91	0.0556	
wave2	1.18e+04	1.64	0.1015	
party_closeAfD:issueeast_germans	1.11e+04	-1.39	0.1644	
party_closeBSW:issueeast_germans	1.11e+04	-0.50	0.6159	
party_closeCDU/CSU:issueeast_germans	1.11e+04	2.36	0.0185	
party_closeFDP:issueeast_germans	1.11e+04	1.14	0.2525	
party_closeGreen Party:issueeast_germans	1.11e+04	-4.83	1.4e-06	
party_closeLeft Party:issueeast_germans	1.11e+04	-7.02	2.3e-12	
party_closeSPD:issueeast_germans	1.11e+04	-1.71	0.0873	
party_closeAfD:issueecon_inequality	1.11e+04	-1.51	0.1311	
party_closeBSW:issueecon_inequality	1.11e+04	-0.56	0.5734	
party_closeCDU/CSU:issueecon_inequality	1.11e+04	0.93	0.3541	
party_closeFDP:issueecon_inequality	1.11e+04	0.18	0.8576	
party_closeGreen Party:issueecon_inequality	1.11e+04	-2.61	0.0091	
party_closeLeft Party:issueecon_inequality	1.11e+04	-1.00	0.3165	
party_closeSPD:issueecon_inequality	1.11e+04	0.38	0.7069	
party_closeAfD:issuegay_marriage	1.11e+04	-3.11	0.0019	
party_closeBSW:issuegay_marriage	1.11e+04	-0.74	0.4583	
party_closeCDU/CSU:issuegay_marriage	1.11e+04	1.52	0.1297	
party_closeFDP:issuegay_marriage	1.11e+04	0.77	0.4430	
party_closeGreen Party:issuegay_marriage	1.11e+04	1.12	0.2614	
party_closeLeft Party:issuegay_marriage	1.11e+04	0.59	0.5556	
party_closeSPD:issuegay_marriage	1.11e+04	0.92	0.3573	
party_closeAfD:issueregulate_internet	1.11e+04	-1.15	0.2489	
party_closeBSW:issueregulate_internet	1.11e+04	-1.70	0.0900	
party_closeCDU/CSU:issueregulate_internet	1.11e+04	0.86	0.3917	
party_closeFDP:issueregulate_internet	1.11e+04	-1.40	0.1616	
party_closeGreen Party:issueregulate_internet	1.11e+04	-4.77	1.9e-06	
party_closeLeft Party:issueregulate_internet	1.11e+04	-6.03	1.7e-09	
party_closeSPD:issueregulate_internet	1.11e+04	-0.92	0.3583	
party_closeAfD:issuerights_indep_integration	1.11e+04	0.84	0.4033	
party_closeBSW:issuerights_indep_integration	1.11e+04	-1.58	0.1133	
party_closeCDU/CSU:issuerights_indep_integration	1.11e+04	0.54	0.5874	
party_closeFDP:issuerights_indep_integration	1.11e+04	1.26	0.2082	
party_closeGreen Party:issuerights_indep_integration	1.11e+04	-3.02	0.0025	
party_closeLeft Party:issuerights_indep_integration	1.11e+04	-2.12	0.0337	

party_closeSPD:issuerights_indep_integration 1.11e+04 -3.29 0.0010

(Intercept) ***
party_closeAfD **
party_closeBSW
party_closeCDU/CSU **
party_closeFDP
party_closeGreen Party ***
party_closeLeft Party ***
party_closeSPD
issueeast_germans ***
issueecon_inequality **
issuegay_marriage *
issueregulate_internet ***
issuerights_indep_integration .
wave2
party_closeAfD:issueeast_germans
party_closeBSW:issueeast_germans
party_closeCDU/CSU:issueeast_germans *
party_closeFDP:issueeast_germans
party_closeGreen Party:issueeast_germans ***
party_closeLeft Party:issueeast_germans ***
party_closeSPD:issueeast_germans .
party_closeAfD:issueecon_inequality
party_closeBSW:issueecon_inequality
party_closeCDU/CSU:issueecon_inequality
party_closeFDP:issueecon_inequality
party_closeGreen Party:issueecon_inequality **
party_closeLeft Party:issueecon_inequality
party_closeSPD:issueecon_inequality
party_closeAfD:issuegay_marriage **
party_closeBSW:issuegay_marriage
party_closeCDU/CSU:issuegay_marriage
party_closeFDP:issuegay_marriage
party_closeGreen Party:issuegay_marriage
party_closeLeft Party:issuegay_marriage
party_closeSPD:issuegay_marriage
party_closeAfD:issueregulate_internet
party_closeBSW:issueregulate_internet .
party_closeCDU/CSU:issueregulate_internet
party_closeFDP:issueregulate_internet
party_closeGreen Party:issueregulate_internet ***
party_closeLeft Party:issueregulate_internet ***
party_closeSPD:issueregulate_internet
party_closeAfD:issuerights_indep_integration
party_closeBSW:issuerights_indep_integration
party_closeCDU/CSU:issuerights_indep_integration
party_closeFDP:issuerights_indep_integration
party_closeGreen Party:issuerights_indep_integration **
party_closeLeft Party:issuerights_indep_integration *
party_closeSPD:issuerights_indep_integration **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Random effects - full model:

Groups	Name	Std.Dev.
id	(Intercept)	0.15313
Residual		0.18394

Verdict - H6

LRT party × issue interaction: $\chi^2(42) = 397.03$, $p = 0.000$
→ Supported: party affiliation significantly predicts issue weights

For exponential/linear weights, the weights are constrained. So there can not be any random effects on the intercept!

```
k <- "exp"  
alpha_cols <- paste0(k, "_alpha_", questions_sc)
```

```

# --- Data (identical) ---
df_h6 <- df_p %>%
  filter(wave %in% c(1, 2)) %>%
  select(id, wave, party_close, all_of(alpha_cols)) %>%
  filter(party_close %in% c(partiesCloseSel, "No party")) %>%
  pivot_longer(
    cols = all_of(alpha_cols),
    names_to = "issue",
    names_prefix = paste0(k, "_alpha_"),
    values_to = "alpha_q"
  ) %>%
  filter(!is.na(party_close), !is.na(alpha_q)) %>%
  mutate(
    party_close = relevel(factor(party_close), ref = "No party"),
    issue = relevel(factor(issue), ref = "climate_concern"),
    wave = factor(wave)
  )

# --- Models ---
# No random intercept: sum-to-1 constraint within person-wave means
# the person-level mean is fixed at 1/k_issues, leaving no between-person
# variance for a random intercept to capture.
# Clustered SEs on id account for within-person correlation across issues/waves.
model_h6_null <- lm_robust(alpha_q ~ issue + wave,
  data = df_h6, clusters = id, se_type = "CR2")
model_h6_full <- lm_robust(alpha_q ~ party_close * issue + wave,
  data = df_h6, clusters = id, se_type = "CR2")

# --- Model comparison via AIC/BIC on plain lm (lm_robust doesn't expose logLik) ---
model_h6_null_lm <- lm(alpha_q ~ issue + wave, data = df_h6)
model_h6_full_lm <- lm(alpha_q ~ party_close * issue + wave, data = df_h6)
wt <- anova(model_h6_null_lm, model_h6_full_lm)

ic_compare <- data.frame(
  Model = c("Null (issue + wave)", "Full (party × issue + wave)"),
  AIC = round(c(AIC(model_h6_null_lm), AIC(model_h6_full_lm)), 1),
  BIC = round(c(BIC(model_h6_null_lm), BIC(model_h6_full_lm)), 1),
  logLik = round(c(as.numeric(logLik(model_h6_null_lm)), as.numeric(logLik(model_h6_full_lm))), 1),
  delta_AIC = round(c(AIC(model_h6_null_lm), AIC(model_h6_full_lm)) - min(c(AIC(model_h6_null_lm), AIC(model_h6_full_lm))), 1),
  delta_BIC = round(c(BIC(model_h6_null_lm), BIC(model_h6_full_lm)) - min(c(BIC(model_h6_null_lm), BIC(model_h6_full_lm))), 1),
  npar = c(length(coef(model_h6_null_lm)), length(coef(model_h6_full_lm)))
)

# R² from plain lm
r2_null <- summary(model_h6_null_lm)$r.squared
r2_full <- summary(model_h6_full_lm)$r.squared

# --- Output ---
cat("=== H6: Party preference predicts issue weights (exponential) ===\n")
cat(" (sum-to-1 constraint → lm with clustered SEs on id; wave as fixed effect)\n")
cat("Model fit:\n")
print(ic_compare, row.names = FALSE)
cat(sprintf(
  "\nR²\n Null : %.3f\n Full : %.3f\n",
  r2_null, r2_full
))
cat("\nLikelihood ratio test (null vs. full):\n")
print(wt)
cat("\nFixed effects - full model:\n")
printCoefmat(summary(model_h6_full)$coefficients[, c("Estimate", "Std. Error", "t value", "Pr(>|t|)"),
  digits = 3, signif.stars = TRUE)
cat("\nVerdict - H6 (exponential)\n")
p_val <- wt$`Pr(>F)`[2]
cat(sprintf(

```

```
" F-test party × issue interaction: F(%d, %d) = %.2f, p = %.3f\n",
  wt$Df[2], wt$Res.Df[2], wt$F[2], p_val
))
if (p_val < .05) {
  cat(" → Supported: party affiliation significantly predicts issue weights\n")
} else {
  cat(" → Not supported: no significant effect of party on issue weights\n")
}
```

=== H6: Party preference predicts issue weights (exponential) ===
(sum-to-1 constraint → lm with clustered SEs on id; wave as fixed effect)
Model fit:

	Model	AIC	BIC	logLik	delta_AIC	delta_BIC	npar
Null (issue + wave)		-2225.7	-2166.3	1120.9	0	0	7
Full (party × issue + wave)		-2213.2	-1842.0	1156.6	12	324	49

R²
Null : 0.008
Full : 0.014

Likelihood ratio test (null vs. full):
Analysis of Variance Table

Model 1: alpha_q ~ issue + wave
Model 2: alpha_q ~ party_close * issue + wave
Res.Df RSS Df Sum of Sq F Pr(>F)
1 12377 605.02
2 12335 601.54 42 3.4848 1.7014 0.003129 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Fixed effects - full model:

	Estimate	Std. Error
(Intercept)	1.87e-01	1.31e-02
party_closeAfD	-1.08e-02	1.83e-02
party_closeBSW	1.85e-02	3.41e-02
party_closeCDU/CSU	-1.53e-02	1.69e-02
party_closeFDP	-3.12e-02	2.72e-02
party_closeGreen Party	2.28e-02	1.93e-02
party_closeLeft Party	4.00e-02	2.29e-02
party_closeSPD	5.78e-03	1.87e-02
issueeast_germans	-3.21e-02	1.94e-02
issueecon_inequality	-7.04e-03	2.13e-02
issuegay_marriage	-5.03e-02	2.02e-02
issueregulate_internet	-4.92e-02	1.85e-02
issuerights_indep_integration	1.81e-02	2.01e-02
wave2	3.37e-09	2.38e-08
party_closeAfD:issueeast_germans	1.35e-02	2.78e-02
party_closeBSW:issueeast_germans	3.59e-02	5.95e-02
party_closeCDU/CSU:issueeast_germans	3.55e-02	2.60e-02
party_closeFDP:issueeast_germans	5.69e-02	4.24e-02
party_closeGreen Party:issueeast_germans	-1.38e-02	2.84e-02
party_closeLeft Party:issueeast_germans	-8.27e-02	3.09e-02
party_closeSPD:issueeast_germans	1.58e-02	2.92e-02
party_closeAfD:issueecon_inequality	1.99e-02	3.10e-02
party_closeBSW:issueecon_inequality	-2.98e-02	4.80e-02
party_closeCDU/CSU:issueecon_inequality	-1.08e-02	2.72e-02
party_closeFDP:issueecon_inequality	1.42e-03	4.11e-02
party_closeGreen Party:issueecon_inequality	-6.12e-02	2.99e-02
party_closeLeft Party:issueecon_inequality	-6.32e-02	3.42e-02
party_closeSPD:issueecon_inequality	-2.67e-02	2.78e-02
party_closeAfD:issuegay_marriage	1.44e-02	2.78e-02
party_closeBSW:issuegay_marriage	1.09e-03	5.10e-02
party_closeCDU/CSU:issuegay_marriage	5.30e-02	2.65e-02
party_closeFDP:issuegay_marriage	6.80e-02	4.23e-02

party_closeGreen Party:issuegay_marriage	1.52e-02	2.93e-02
party_closeLeft Party:issuegay_marriage	-1.17e-02	3.36e-02
party_closeSPD:issuegay_marriage	2.12e-02	2.91e-02
party_closeAfD:issueregulate_internet	-1.04e-02	2.52e-02
party_closeBSW:issueregulate_internet	-5.81e-02	4.12e-02
party_closeCDU/CSU:issueregulate_internet	1.39e-02	2.39e-02
party_closeFDP:issueregulate_internet	2.77e-02	3.59e-02
party_closeGreen Party:issueregulate_internet	-3.31e-02	2.64e-02
party_closeLeft Party:issueregulate_internet	-3.78e-02	3.16e-02
party_closeSPD:issueregulate_internet	2.04e-03	2.72e-02
party_closeAfD:issuerights_indep_integration	2.75e-02	2.92e-02
party_closeBSW:issuerights_indep_integration	-5.99e-02	5.03e-02
party_closeCDU/CSU:issuerights_indep_integration	2.45e-04	2.66e-02
party_closeFDP:issuerights_indep_integration	3.35e-02	4.82e-02
party_closeGreen Party:issuerights_indep_integration	-4.41e-02	2.90e-02
party_closeLeft Party:issuerights_indep_integration	-4.47e-02	3.81e-02
party_closeSPD:issuerights_indep_integration	-4.70e-02	2.81e-02
	t value	Pr(> t)
(Intercept)	14.28	<2e-16 ***
party_closeAfD	-0.59	0.5551
party_closeBSW	0.54	0.5893
party_closeCDU/CSU	-0.90	0.3661
party_closeFDP	-1.15	0.2548
party_closeGreen Party	1.19	0.2364
party_closeLeft Party	1.75	0.0816 .
party_closeSPD	0.31	0.7572
issueeast_germans	-1.65	0.0995 .
issueecon_inequality	-0.33	0.7415
issuegay_marriage	-2.49	0.0135 *
issueregulate_internet	-2.66	0.0084 **
issuerights_indep_integration	0.90	0.3693
wave2	0.14	0.8876
party_closeAfD:issueeast_germans	0.49	0.6264
party_closeBSW:issueeast_germans	0.60	0.5492
party_closeCDU/CSU:issueeast_germans	1.36	0.1734
party_closeFDP:issueeast_germans	1.34	0.1842
party_closeGreen Party:issueeast_germans	-0.49	0.6268
party_closeLeft Party:issueeast_germans	-2.68	0.0080 **
party_closeSPD:issueeast_germans	0.54	0.5880
party_closeAfD:issueecon_inequality	0.64	0.5203
party_closeBSW:issueecon_inequality	-0.62	0.5370
party_closeCDU/CSU:issueecon_inequality	-0.40	0.6923
party_closeFDP:issueecon_inequality	0.03	0.9726
party_closeGreen Party:issueecon_inequality	-2.05	0.0414 *
party_closeLeft Party:issueecon_inequality	-1.85	0.0657 .
party_closeSPD:issueecon_inequality	-0.96	0.3375
party_closeAfD:issuegay_marriage	0.52	0.6054
party_closeBSW:issuegay_marriage	0.02	0.9830
party_closeCDU/CSU:issuegay_marriage	2.00	0.0460 *
party_closeFDP:issuegay_marriage	1.61	0.1127
party_closeGreen Party:issuegay_marriage	0.52	0.6038
party_closeLeft Party:issuegay_marriage	-0.35	0.7271
party_closeSPD:issuegay_marriage	0.73	0.4662
party_closeAfD:issueregulate_internet	-0.41	0.6795
party_closeBSW:issueregulate_internet	-1.41	0.1627
party_closeCDU/CSU:issueregulate_internet	0.58	0.5608
party_closeFDP:issueregulate_internet	0.77	0.4422
party_closeGreen Party:issueregulate_internet	-1.25	0.2112
party_closeLeft Party:issueregulate_internet	-1.20	0.2330
party_closeSPD:issueregulate_internet	0.08	0.9402
party_closeAfD:issuerights_indep_integration	0.94	0.3463
party_closeBSW:issuerights_indep_integration	-1.19	0.2381
party_closeCDU/CSU:issuerights_indep_integration	0.01	0.9926
party_closeFDP:issuerights_indep_integration	0.69	0.4900
party_closeGreen Party:issuerights_indep_integration	-1.52	0.1291

```

party_closeLeft Party:issuerights_indep_integration    -1.17    0.2420
party_closeSPD:issuerights_indep_integration          -1.68    0.0946 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Verdict - H6 (exponential)

F-test party × issue interaction: F(42, 12335) = 1.70, p = 0.003

→ Supported: party affiliation significantly predicts issue weights

3.2.1.3 Interpretation 2: For each issue, does party matter?

```

k      <- "corrS"
alpha_cols <- paste0(k, "_alpha_", questions_sc)

cat("  H6 | Interpretation 2: Party effects per issue (Waves 1-2)\n")
cat("  LMM with HC3 robust SEs; reference = 'No party'\n")
cat("  Fixed: party + wave | Random: intercept per id\n")

results_list <- vector("list", length(questions_sc))

for (i in seq_along(questions_sc)) {
  q <- questions_sc[i]

  df_q <- df_p %>%
    filter(wave %in% c(1, 2)) %>%
    select(id, wave, party_close, alpha_q = all_of(paste0(k, "_alpha_", q))) %>%
    filter(party_close %in% c(partiesCloseSel, "No party")) %>%
    mutate(
      party_close = relevel(factor(party_close), ref = "No party"),
      wave        = factor(wave)
    ) %>%
    filter(!is.na(alpha_q))

  m      <- lmer(alpha_q ~ party_close + wave + (1 | id), data = df_q)
  r2_vals <- r2(m)
  n_val  <- nobs(m)
  n_id   <- length(unique(df_q$id))

  # Omnibus test for party via LRT (null without party_close)
  m_null <- lmer(alpha_q ~ wave + (1 | id), data = df_q)
  wt_q   <- anova(m_null, m)
  p_omni <- wt_q$`Pr(>Chisq)`[2]
  chi_omni <- wt_q$Chisq[2]
  df_omni <- wt_q$Df[2]

  sig_omni <- case_when(
    p_omni < 0.001 ~ "***",
    p_omni < 0.01  ~ "**",
    p_omni < 0.05  ~ "*",
    p_omni < 0.1   ~ ".",
    TRUE          ~ "ns"
  )

  coef_raw <- summary(m)$coefficients

  coef_df <- as.data.frame(coef_raw) %>%
    setNames(c("Estimate", "SE", "df", "t", "Pr(>|t|)")) %>% # lmer gives no p-column by default;
    # lmerTest adds it - adjust if needed

    mutate(
      Party    = gsub("party_close", "", rownames(coef_raw)),
      p        = 2 * pt(-abs(t), df = n_id - 1), # Satterthwaite already in lmerTest
      sig      = case_when(
        p < 0.001 ~ "***",
        p < 0.01  ~ "**",
        p < 0.05  ~ "*"
      )
    )
}

```

```
p < 0.1 ~ ".",
TRUE ~ ""
),
across(c(Estimate, SE), ~round(., 3)),
t = round(t, 2),
p = round(p, 3)
) %>%
select(Party, Estimate, SE, t, p, sig)
rownames(coef_df) <- NULL

cat(sprintf(
  "\nIssue : %s\n N = %d obs / %d persons | R² marg/cond = %.3f / %.3f | LRT ²(%d) = %.2f, p = %.4f %s",q, n,
  r2_vals$R2_marginal, r2_vals$R2_conditional,
  df_omni, chi_omni, p_omni, sig_omni
))
print(kable(coef_df, format = "simple", align = c("l", "r", "r", "r", "r", "l")))
}
```

refitting model(s) with ML (instead of REML)
refitting model(s) with ML (instead of REML)
refitting model(s) with ML (instead of REML)
refitting model(s) with ML (instead of REML)
refitting model(s) with ML (instead of REML)
refitting model(s) with ML (instead of REML)

H6 | Interpretation 2: Party effects per issue (Waves 1-2)
LMM with HC3 robust SEs; reference = 'No party'
Fixed: party + wave | Random: intercept per id

Issue : climate_concern
N = 2062 obs / 1195 persons | R² marg/cond = 0.066 / 0.566 | LRT ²(7) = 103.31, p = 0.0000 ***

Party	Estimate	SE	t	p	sig
(Intercept)	0.273	0.015	18.00	0.000	***
AfD	0.059	0.020	2.89	0.004	**
BSW	0.006	0.034	0.18	0.857	
CDU/CSU	-0.044	0.020	-2.24	0.025	*
FDP	-0.020	0.034	-0.59	0.553	
Green Party	0.134	0.023	5.94	0.000	***
Left Party	0.143	0.026	5.54	0.000	***
SPD	0.000	0.021	0.00	0.997	
wave2	0.012	0.008	1.50	0.133	

Issue : gay_marriage
N = 2063 obs / 1195 persons | R² marg/cond = 0.075 / 0.529 | LRT ²(7) = 120.27, p = 0.0000 ***

Party	Estimate	SE	t	p	sig
(Intercept)	0.248	0.015	16.77	0.000	***
AfD	0.009	0.020	0.44	0.662	
BSW	-0.032	0.033	-0.97	0.332	
CDU/CSU	-0.030	0.019	-1.56	0.120	
FDP	-0.010	0.034	-0.29	0.774	
Green Party	0.154	0.022	7.06	0.000	***
Left Party	0.158	0.025	6.29	0.000	***
SPD	0.019	0.021	0.90	0.370	
wave2	0.000	0.008	0.01	0.991	

Issue : rights_indep_integration
N = 2059 obs / 1195 persons | R² marg/cond = 0.049 / 0.542 | LRT ²(7) = 77.51, p = 0.0000 ***

Party	Estimate	SE	t	p	sig
(Intercept)	0.297	0.015	19.27	0.000	***

AfD	0.079	0.021	3.77	0.000	***
BSW	-0.047	0.035	-1.36	0.173	
CDU/CSU	-0.032	0.020	-1.61	0.107	
FDP	0.020	0.035	0.57	0.568	
Green Party	0.073	0.023	3.19	0.001	**
Left Party	0.100	0.026	3.80	0.000	***
SPD	-0.055	0.022	-2.52	0.012	*
wave2	0.008	0.008	0.96	0.339	

Issue : econ_inequality
 N = 2061 obs / 1194 persons | R² marg/cond = 0.035 / 0.480 | LRT ²(7) = 56.93, p = 0.0000 ***

Party	Estimate	SE	t	p	sig
-----	-----	-----	-----	-----	-----
(Intercept)	0.232	0.015	15.81	0.000	***
AfD	0.032	0.020	1.62	0.106	
BSW	-0.017	0.033	-0.53	0.595	
CDU/CSU	-0.032	0.019	-1.70	0.089	.
FDP	-0.016	0.033	-0.50	0.620	
Green Party	0.079	0.022	3.67	0.000	***
Left Party	0.118	0.025	4.76	0.000	***
SPD	0.006	0.020	0.31	0.755	
wave2	0.004	0.008	0.48	0.633	

Issue : regulate_internet
 N = 2062 obs / 1196 persons | R² marg/cond = 0.023 / 0.413 | LRT ²(7) = 39.52, p = 0.0000 ***

Party	Estimate	SE	t	p	sig
-----	-----	-----	-----	-----	-----
(Intercept)	0.209	0.014	14.87	0.000	***
AfD	0.037	0.019	1.97	0.049	*
BSW	-0.080	0.031	-2.55	0.011	*
CDU/CSU	-0.045	0.018	-2.51	0.012	*
FDP	-0.074	0.032	-2.32	0.021	*
Green Party	0.025	0.020	1.21	0.228	
Left Party	-0.001	0.024	-0.06	0.953	
SPD	-0.032	0.020	-1.63	0.104	
wave2	0.002	0.008	0.20	0.839	

Issue : east_germans
 N = 2060 obs / 1194 persons | R² marg/cond = 0.014 / 0.436 | LRT ²(7) = 22.49, p = 0.0021 **

Party	Estimate	SE	t	p	sig
-----	-----	-----	-----	-----	-----
(Intercept)	0.213	0.014	15.64	0.000	***
AfD	0.035	0.018	1.91	0.056	.
BSW	-0.032	0.031	-1.05	0.293	
CDU/CSU	-0.013	0.018	-0.73	0.464	
FDP	0.016	0.031	0.50	0.615	
Green Party	0.029	0.020	1.44	0.149	
Left Party	-0.024	0.023	-1.03	0.305	
SPD	-0.038	0.019	-1.99	0.047	*
wave2	0.010	0.008	1.36	0.174	

And for exponential/linear weights (constrained):

```
k <- "exp"
alpha_cols <- paste0(k, "_alpha_", questions_sc)
cat(" H6 | Interpretation 2: Party effects per issue (Waves 1-2)\n")
cat(" Clustered SEs (CR2) on id; reference = 'No party'\n")
cat(" Fixed: party + wave | No random intercept (sum-to-1 constraint)\n")

results_list <- vector("list", length(questions_sc))
for (i in seq_along(questions_sc)) {
  q <- questions_sc[i]
  df_q <- df_p %>%
```

```

filter(wave %in% c(1, 2)) %>%
select(id, wave, party_close, alpha_q = all_of(paste0(k, "_alpha_", q))) %>%
filter(party_close %in% c(partiesCloseSel, "No party")) %>%
mutate(
  party_close = relevel(factor(party_close), ref = "No party"),
  wave        = factor(wave)
) %>%
filter(!is.na(alpha_q))

# Main model with clustered SEs
m      <- lm_robust(alpha_q ~ party_close + wave, data = df_q, clusters = id, se_type = "CR2")
# Plain lm for R2, AIC, and F-test
m_lm   <- lm(alpha_q ~ party_close + wave, data = df_q)
m_null_lm <- lm(alpha_q ~ wave, data = df_q)

r2_val <- summary(m_lm)$r.squared
n_val  <- nobs(m_lm)
n_id   <- length(unique(df_q$id))

# Omnibus F-test for party via anova on plain lm
wt_q    <- anova(m_null_lm, m_lm)
p_omni  <- wt_q$`Pr(>F)`[2]
f_omni  <- wt_q$F[2]
df1_omni <- wt_q$Df[2]
df2_omni <- wt_q$Res.Df[2]
sig_omni <- case_when(
  p_omni < 0.001 ~ "****",
  p_omni < 0.01  ~ "***",
  p_omni < 0.05  ~ "**",
  p_omni < 0.1   ~ ".",
  TRUE      ~ "ns"
)

# Coefficients from lm_robust (has proper clustered SEs)
coef_raw <- summary(m)$coefficients
coef_df <- as.data.frame(coef_raw) %>%
  setNames(c("Estimate", "SE", "t", "p", "ci_low", "ci_high", "df")) %>%
  mutate(
    Party = gsub("party_close", "", rownames(coef_raw)),
    sig    = case_when(
      p < 0.001 ~ "****",
      p < 0.01  ~ "***",
      p < 0.05  ~ "**",
      p < 0.1   ~ ".",
      TRUE      ~ ""
    ),
    across(c(Estimate, SE), ~round(., 3)),
    t = round(t, 2),
    p = round(p, 3)
  ) %>%
  select(Party, Estimate, SE, t, p, sig)
rownames(coef_df) <- NULL

cat(sprintf(
  "\n Issue : %s\n N = %d obs / %d persons | R2 = %.3f | F(%d, %d) = %.2f, p = %.4f %s",
  q, n_val, n_id,
  r2_val,
  df1_omni, df2_omni, f_omni, p_omni, sig_omni
))
print(kable(coef_df, format = "simple", align = c("l", "r", "r", "r", "r", "l")))
}

```

H6 | Interpretation 2: Party effects per issue (Waves 1-2)
 Clustered SEs (CR2) on id; reference = 'No party'
 Fixed: party + wave | No random intercept (sum-to-1 constraint)

Issue : climate_concern

N = 2064 obs / 1196 persons | $R^2 = 0.006$ | $F(7, 2055) = 1.80$, $p = 0.0831$.

Party	Estimate	SE	t	p	sig
-----	-----	-----	-----	-----	-----
(Intercept)	0.190	0.014	13.58	0.000	***
AfD	-0.011	0.018	-0.58	0.560	
BSW	0.018	0.034	0.54	0.591	
CDU/CSU	-0.016	0.017	-0.92	0.359	
FDP	-0.031	0.027	-1.16	0.252	
Green Party	0.023	0.019	1.18	0.237	
Left Party	0.040	0.023	1.75	0.082	.
SPD	0.006	0.019	0.30	0.763	
wave2	-0.006	0.010	-0.65	0.516	

Issue : gay_marriage

N = 2064 obs / 1196 persons | $R^2 = 0.006$ | $F(7, 2055) = 1.37$, $p = 0.2119$ ns

Party	Estimate	SE	t	p	sig
-----	-----	-----	-----	-----	-----
(Intercept)	0.143	0.013	11.14	0.000	***
AfD	0.004	0.017	0.23	0.819	
BSW	0.019	0.033	0.60	0.553	
CDU/CSU	0.037	0.017	2.25	0.025	*
FDP	0.036	0.027	1.34	0.185	
Green Party	0.038	0.017	2.20	0.029	*
Left Party	0.028	0.019	1.45	0.148	
SPD	0.027	0.018	1.49	0.136	
wave2	-0.015	0.009	-1.54	0.125	

Issue : rights_indep_integration

N = 2064 obs / 1196 persons | $R^2 = 0.007$ | $F(7, 2055) = 1.94$, $p = 0.0600$.

Party	Estimate	SE	t	p	sig
-----	-----	-----	-----	-----	-----
(Intercept)	0.202	0.014	14.68	0.000	***
AfD	0.017	0.020	0.83	0.409	
BSW	-0.041	0.030	-1.36	0.178	
CDU/CSU	-0.015	0.017	-0.85	0.394	
FDP	0.002	0.032	0.07	0.942	
Green Party	-0.021	0.019	-1.14	0.254	
Left Party	-0.005	0.023	-0.20	0.844	
SPD	-0.041	0.018	-2.30	0.022	*
wave2	0.006	0.010	0.58	0.563	

Issue : econ_inequality

N = 2064 obs / 1196 persons | $R^2 = 0.005$ | $F(7, 2055) = 1.56$, $p = 0.1444$ ns

Party	Estimate	SE	t	p	sig
-----	-----	-----	-----	-----	-----
(Intercept)	0.179	0.015	11.89	0.000	***
AfD	0.009	0.021	0.43	0.664	
BSW	-0.011	0.028	-0.41	0.684	
CDU/CSU	-0.026	0.018	-1.45	0.148	
FDP	-0.030	0.026	-1.15	0.255	
Green Party	-0.038	0.018	-2.11	0.036	*
Left Party	-0.023	0.022	-1.07	0.285	
SPD	-0.021	0.018	-1.17	0.243	
wave2	0.002	0.009	0.20	0.839	

Issue : regulate_internet

N = 2064 obs / 1196 persons | $R^2 = 0.003$ | $F(7, 2055) = 0.99$, $p = 0.4333$ ns

Party	Estimate	SE	t	p	sig
-----	-----	-----	-----	-----	-----

(Intercept)	0.138	0.011	12.17	0.000	***
AfD	-0.021	0.014	-1.49	0.138	
BSW	-0.040	0.021	-1.91	0.060	.
CDU/CSU	-0.001	0.015	-0.10	0.922	
FDP	-0.004	0.022	-0.16	0.874	
Green Party	-0.010	0.015	-0.67	0.504	
Left Party	0.002	0.018	0.12	0.901	
SPD	0.008	0.016	0.49	0.624	
wave2	-0.001	0.008	-0.13	0.893	

Issue : east_germans
N = 2064 obs / 1196 persons | R² = 0.009 | F(7, 2055) = 2.34, p = 0.0222 *

Party	Estimate	SE	t	p	sig
(Intercept)	0.148	0.013	11.53	0.000	***
AfD	0.002	0.018	0.14	0.892	
BSW	0.054	0.037	1.47	0.146	
CDU/CSU	0.021	0.017	1.23	0.220	
FDP	0.026	0.028	0.94	0.352	
Green Party	0.009	0.017	0.53	0.594	
Left Party	-0.042	0.018	-2.36	0.019	*
SPD	0.022	0.019	1.18	0.238	
wave2	0.015	0.009	1.60	0.110	

3.2.2 H7

3.2.2.1 H7. Left-Right placement predicts issue weights.

We model the issue weights for each issue q as

$$\alpha_{n,q} = 1/6 + \beta_q \cdot lr_n + u_q + \epsilon_{n,q} \tag{8}$$

with continuous variable lr_n between **left'** (\$-5\$) and **right'** (5). We estimate this model jointly across all six issues. % β_0 is average. Because α is normalised: $\beta_0 = 1/6$ % lr_n is explanatory variable % u_q is average deviation of the weight for issue q % $\epsilon_{\{nq\}}$ is individual specific error for this question and this individual.

The prediction is $\beta_q \neq 0$. Left and right participants differ in how strongly they weigh different political issues when mapping individuals. % We test this via a single F-test on the joint significance of $\{\beta_q\}$, with significance threshold $\alpha = 0.05$.

```
k = "exp"

models_h7 <- lapply(setNames(questions_sc, questions_sc), function(q) {
  df_q <- df_p %>%
    filter(wave %in% c(1,2)) %>%
    select(id, wave, lr, alpha_q = all_of(paste0(k, "_alpha_", q))) %>%
    filter(!is.na(alpha_q))

  lmer(alpha_q ~ lr + (1|id) + wave, data = df_q)
})

# --- Report ---
for (q in questions_sc) {
  m <- models_h7[[q]]
  coef_df <- as.data.frame(summary(m)$coefficients)

  est <- coef_df["lr", "Estimate"]
  pval <- coef_df["lr", "Pr(>|t|)"]

  cat(sprintf(
    "%s  alpha = %.3f + %.3f * lr  (p = %.4f, %s)\n",
    q, coef_df["(Intercept)", "Estimate"], est, pval,
    ifelse(pval < 0.05, "reject H0: TRUE", "fail to reject H0: FALSE")
  ))
}
```

```
climate_concern  alpha = 0.215 + -0.053 * lr  (p = 0.0307, reject H0: TRUE)
gay_marriage     alpha = 0.192 + -0.052 * lr  (p = 0.0322, reject H0: TRUE)
```

rights_indep_integration alpha = 0.180 + 0.017 * lr (p = 0.4958, fail to reject H0: FALSE)
econ_inequality alpha = 0.144 + 0.035 * lr (p = 0.1400, fail to reject H0: FALSE)
regulate_internet alpha = 0.125 + 0.023 * lr (p = 0.2768, fail to reject H0: FALSE)
east_germans alpha = 0.144 + 0.028 * lr (p = 0.2374, fail to reject H0: FALSE)

Interpretation 2: Joint Model

```
# long format
df_long <- df_p %>%
  filter(wave %in% c(1,2)) %>%
  select(id, wave, lr,
         starts_with(paste0(k, "_alpha_"))) %>%
  pivot_longer(
    cols = starts_with(paste0(k, "_alpha_")),
    names_to = "issue",
    values_to = "alpha"
  ) %>%
  mutate(
    issue = gsub(paste0(k, "_alpha_"), "", issue),
    issue = factor(issue)
  ) %>%
  filter(!is.na(alpha))

# full model: separate intercepts and separate LR slopes by issue
m_full <- lmer(
  alpha ~ issue + issue:lr + wave + (1 | id),
  data = df_long
)
```

boundary (singular) fit: see help('isSingular')

```
m_null <- lmer(alpha ~ issue + wave + (1|id), data = df_long)
```

boundary (singular) fit: see help('isSingular')

```
summary(m_full)
```

Correlation matrix not shown by default, as p = 13 > 12.
Use print(x, correlation=TRUE) or
vcov(x) if you need it

```
anova(m_null, m_full)
```

refitting model(s) with ML (instead of REML)

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: alpha ~ issue + issue:lr + wave + (1 | id)
Data: df_long

REML criterion at convergence: -2198.7

Scaled residuals:

Min	1Q	Median	3Q	Max
-0.9668	-0.7180	-0.4186	0.4067	3.9368

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.00000	0.0000
Residual		0.04897	0.2213

Number of obs: 12774, groups: id, 1227

Fixed effects:

	Estimate	Std. Error	df	t value
(Intercept)	2.139e-01	1.222e-02	1.276e+04	17.505
issueeast_germans	-6.443e-02	1.712e-02	1.276e+04	-3.764
issueecon_inequality	-6.898e-02	1.712e-02	1.276e+04	-4.030
issuegay_marriage	-2.826e-02	1.712e-02	1.276e+04	-1.651

```

issueregulate_internet      -9.129e-02  1.712e-02  1.276e+04  -5.334
issuerights_indep_integration -3.071e-02  1.712e-02  1.276e+04  -1.794
wave2                       2.610e-09  3.935e-03  1.276e+04   0.000
issueclimate_concern:lr     -5.287e-02  2.277e-02  1.276e+04  -2.322
issueeast_germans:lr        2.951e-02  2.277e-02  1.276e+04   1.296
issueecon_inequality:lr     3.711e-02  2.277e-02  1.276e+04   1.630
issuegay_marriage:lr        -5.361e-02  2.277e-02  1.276e+04  -2.354
issueregulate_internet:lr   2.201e-02  2.277e-02  1.276e+04   0.967
issuerights_indep_integration:lr 1.786e-02  2.277e-02  1.276e+04   0.784

```

```

Pr(>|t|)
(Intercept)                < 2e-16 ***
issueeast_germans           0.000168 ***
issueecon_inequality        5.61e-05 ***
issuegay_marriage           0.098805 .
issueregulate_internet      9.78e-08 ***
issuerights_indep_integration 0.072820 .
wave2                       0.999999
issueclimate_concern:lr     0.020246 *
issueeast_germans:lr        0.195020
issueecon_inequality:lr     0.103201
issuegay_marriage:lr        0.018571 *
issueregulate_internet:lr   0.333786
issuerights_indep_integration:lr 0.432904
---

```

```

Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
optimizer (nloptwrap) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

```

Data: df_long

Models:

```

m_null: alpha ~ issue + wave + (1 | id)
m_full: alpha ~ issue + issue:lr + wave + (1 | id)
      npar    AIC      BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
m_null    9 -2260.6 -2193.5 1139.3  -2278.6
m_full   15 -2265.4 -2153.6 1147.7  -2295.4 16.826  6   0.009944 **
---

```

```

Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

3.2.3 H8

3.2.3.1 H8. Age predicts issue weights.

We model the issue weights for each issue q as

$$\alpha_{n,q} = 1/6 + \beta_q \cdot age_n + u_q + \epsilon_{n,q} \quad (9)$$

The prediction is $\beta_q \neq 0$. Older and younger participants differ in how strongly they weigh different political issues when mapping individuals.

```

k = "exp"

models_h8 <- lapply(setNames(questions_sc, questions_sc), function(q) {
  df_q <- df_p %>%
    filter(wave %in% c(1,2)) %>%
    select(id, wave, age, alpha_q = all_of(paste0(k, "_alpha_", q))) %>%
    filter(!is.na(alpha_q), !is.na(age)) %>%
    mutate(age_z = as.numeric(scale(age)))

  lmer(alpha_q ~ age_z + (1|id) + wave, data = df_q, REML = FALSE)
})

# --- Report ---
for (q in questions_sc) {
  m <- models_h8[[q]]
  coef_df <- as.data.frame(summary(m)$coefficients)

  est <- coef_df["age_z", "Estimate"]
  pval <- coef_df["age_z", "Pr(>|t|)"]
}

```

```
cat(sprintf(
  "%s  alpha = %.3f + %.3f * age_z  (p = %.4f, %s)\n",
  q, coef_df["(Intercept)", "Estimate"], est, pval,
  ifelse(pval < 0.05, "reject H0: TRUE", "fail to reject H0: FALSE")
))
}
```

```
climate_concern  alpha = 0.190 + 0.014 * age_z  (p = 0.0055, reject H0: TRUE)
gay_marriage     alpha = 0.167 + -0.007 * age_z  (p = 0.1689, fail to reject H0: FALSE)
rights_indep_integration  alpha = 0.186 + -0.008 * age_z  (p = 0.1164, fail to reject H0: FALSE)
econ_inequality  alpha = 0.162 + -0.007 * age_z  (p = 0.1425, fail to reject H0: FALSE)
regulate_internet  alpha = 0.137 + 0.005 * age_z  (p = 0.2480, fail to reject H0: FALSE)
east_germans      alpha = 0.158 + 0.003 * age_z  (p = 0.5024, fail to reject H0: FALSE)
```

Interpretation 2: Joint Model

```
# long format
df_long <- df_p %>%
  filter(wave %in% c(1,2)) %>%
  filter(!is.na(age)) %>%
  mutate(age_z = as.numeric(scale(age))) %>%
  select(id, wave, age_z,
         starts_with(paste0(k, "_alpha_"))) %>%
  pivot_longer(
    cols = starts_with(paste0(k, "_alpha_")),
    names_to = "issue",
    values_to = "alpha"
  ) %>%
  mutate(
    issue = gsub(paste0(k, "_alpha_"), "", issue),
    issue = factor(issue)
  ) %>%
  filter(!is.na(alpha))

# full model: separate intercepts and separate LR slopes by issue
m_full <- lmer(
  alpha ~ issue + issue:age_z + wave + (1 | id),
  data = df_long, REML = FALSE
)
```

boundary (singular) fit: see help('isSingular')

```
m_null <- lmer(alpha ~ issue + wave + (1|id), data = df_long, REML = FALSE)
```

boundary (singular) fit: see help('isSingular')

```
summary(m_full)
```

Correlation matrix not shown by default, as p = 13 > 12.

Use print(x, correlation=TRUE) or

vcov(x) if you need it

```
anova(m_null, m_full)
```

Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's

method [lmerModLmerTest]

Formula: alpha ~ issue + issue:age_z + wave + (1 | id)

Data: df_long

AIC	BIC	logLik	-2*log(L)	df.resid
-2237.3	-2125.6	1133.6	-2267.3	12657

Scaled residuals:

Min	1Q	Median	3Q	Max
-0.9632	-0.7180	-0.4194	0.4060	3.9468

Random effects:

```
Groups   Name      Variance Std.Dev.
id       (Intercept) 0.00000  0.0000
Residual                0.04896  0.2213
Number of obs: 12672, groups: id, 1216

Fixed effects:
              Estimate Std. Error      df t value
(Intercept)      1.888e-01  5.135e-03 1.267e+04  36.764
issueeast_germans -2.478e-02  6.809e-03 1.267e+04  -3.640
issueecon_inequality -2.561e-02  6.809e-03 1.267e+04  -3.761
issuegay_marriage  -2.951e-02  6.809e-03 1.267e+04  -4.334
issueregulate_internet -5.486e-02  6.809e-03 1.267e+04  -8.057
issuerights_indep_integration 2.153e-03  6.809e-03 1.267e+04   0.316
wave2            2.898e-09  3.950e-03 1.267e+04   0.000
issueclimate_concern:age_z  1.449e-02  4.816e-03 1.267e+04   3.009
issueeast_germans:age_z    3.322e-03  4.816e-03 1.267e+04   0.690
issueecon_inequality:age_z -7.587e-03  4.816e-03 1.267e+04  -1.575
issuegay_marriage:age_z   -7.140e-03  4.816e-03 1.267e+04  -1.483
issueregulate_internet:age_z  5.122e-03  4.816e-03 1.267e+04   1.064
issuerights_indep_integration:age_z -8.206e-03  4.816e-03 1.267e+04  -1.704
Pr(>|t|)
(Intercept)      < 2e-16 ***
issueeast_germans  0.000274 ***
issueecon_inequality 0.000170 ***
issuegay_marriage  1.48e-05 ***
issueregulate_internet 8.50e-16 ***
issuerights_indep_integration 0.751817
wave2            0.999999
issueclimate_concern:age_z 0.002629 **
issueeast_germans:age_z    0.490377
issueecon_inequality:age_z 0.115194
issuegay_marriage:age_z   0.138217
issueregulate_internet:age_z 0.287570
issuerights_indep_integration:age_z 0.088400 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
optimizer (nloptwrap) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')
```

```
Data: df_long
Models:
m_null: alpha ~ issue + wave + (1 | id)
m_full: alpha ~ issue + issue:age_z + wave + (1 | id)
      npar      AIC      BIC logLik -2*log(L) Chisq Df Pr(>Chisq)
m_null    9 -2231.1 -2164.0 1124.5  -2249.1
m_full   15 -2237.3 -2125.6 1133.7  -2267.3 18.23  6  0.005683 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

3.2.4 Exploration

3.2.4.1 Explorative Analyses.

We also model **H6** to **H8** with different combinations of party preference, left-right placement, and age of the participants to model participants issue weights $\alpha_{n,q}$. In addition, we model how political interest, pi_n , pf_n , interacts with the explanatory variables above.

```
print(" exploration later ")
```

```
[1] " exploration later "
```

3.3 Aim 3

Aim 3: Assess the role of social circle belief variation

For Aim 3, we test whether including social circle belief variation improves predictions for participants’ issue weights (inferred from the map distances), their reported issue importance, likeability of voters, and polarisation perceptions. In addition, we explore how social circle belief improves predictions of the map distances between individuals.

3.3.1 H9

3.3.1.1 H9. Social circle variance predicts issue weights.

We model issue weights of participant n as

$$\alpha_{n,q} = 1/6 + \beta_q \cdot \text{var}_{n,q} + u_q + \epsilon_{n,q} . \tag{10}$$

The prediction is $\beta_q < 0$. Issues are assigned higher weight when associated with lower social circle variance.

{CHANGE TO PREREG: I am using standard deviation instead of variance now!!!}

```
variance_type = "std"
k = "exp"

df_h9 <- df_p |>
  filter(wave %in% c(1, 2)) |>
  pivot_longer(
    cols      = matches(paste0("^(", variance_type, "_socialCircle_ops|", k, "_alpha)_")),
    names_to   = c(".value", "issue"),
    names_pattern = paste0("^(", variance_type, "_socialCircle_ops|", k, "_alpha)_(.*)")
  ) |>
  rename(
    var_sc = paste0(variance_type, "_socialCircle_ops"),
    alpha = paste0(k, "_alpha")
  ) |>
  filter(!is.na(alpha), !is.na(var_sc))

model_h9 <- lmer(alpha ~ var_sc + wave + (1 | issue) , data = df_h9)

cat(sprintf("=== H9: social circle %s predicts weights (%s) ===\n", variance_type, k) )

summary(model_h9)

# One-sided p-value for beta < 0
coef_row <- summary(model_h9)$coefficients["var_sc", ]
df_resid <- nrow(df_h9) - length(fixef(model_h9))
t_stat <- coef_row["t value"]
p_h9      <- if (t_stat < 0) pt(t_stat, df_resid) else pt(t_stat, df_resid, lower.tail = FALSE)

coefs <- summary(model_h9)$coefficients
tbl_h9 <- data.frame(
  Term      = c("Intercept", paste0("Social circle ",variance_type), "Wave"),
  Estimate  = round(coefs[, "Estimate"], 3),
  SE        = round(coefs[, "Std. Error"], 3),
  t         = round(coefs[, "t value"], 3),
  row.names = NULL
)

cat(sprintf("=== H9: social circle %s predicts weights (%s) ===\n", variance_type, k))
print(kable(tbl_h9, format = "simple", align = c("l", "r", "r", "r")))
cat(sprintf(
  "\np (one-sided, H1: beta < 0) = %.4f\nDirection correct (beta < 0): %s | Reject H0 at alpha = 0.05: %s\n",
  p_h9,
  ifelse(coef_row["Estimate"] < 0, "Yes", "No"),
  ifelse(coef_row["Estimate"] < 0 & p_h9 < 0.05, "Yes", "No")
))
```

=== H9: social circle std predicts weights (exp) ===
Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]
Formula: alpha ~ var_sc + wave + (1 | issue)
Data: df_h9

REML criterion at convergence: -2295.3

Scaled residuals:
Min 1Q Median 3Q Max

-0.9777 -0.7130 -0.4072 0.4123 4.1382

Random effects:

Groups	Name	Variance	Std.Dev.
issue	(Intercept)	0.0004709	0.0217
Residual		0.0487689	0.2208

Number of obs: 12774, groups: issue, 6

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	1.883e-01	9.631e-03	6.350e+00	19.555	6.50e-07 ***
var_sc	-7.880e-02	9.845e-03	1.275e+04	-8.004	1.31e-15 ***
wave2	1.355e-03	3.930e-03	1.277e+04	0.345	0.73

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	var_sc
var_sc	-0.281	
wave2	-0.172	-0.043

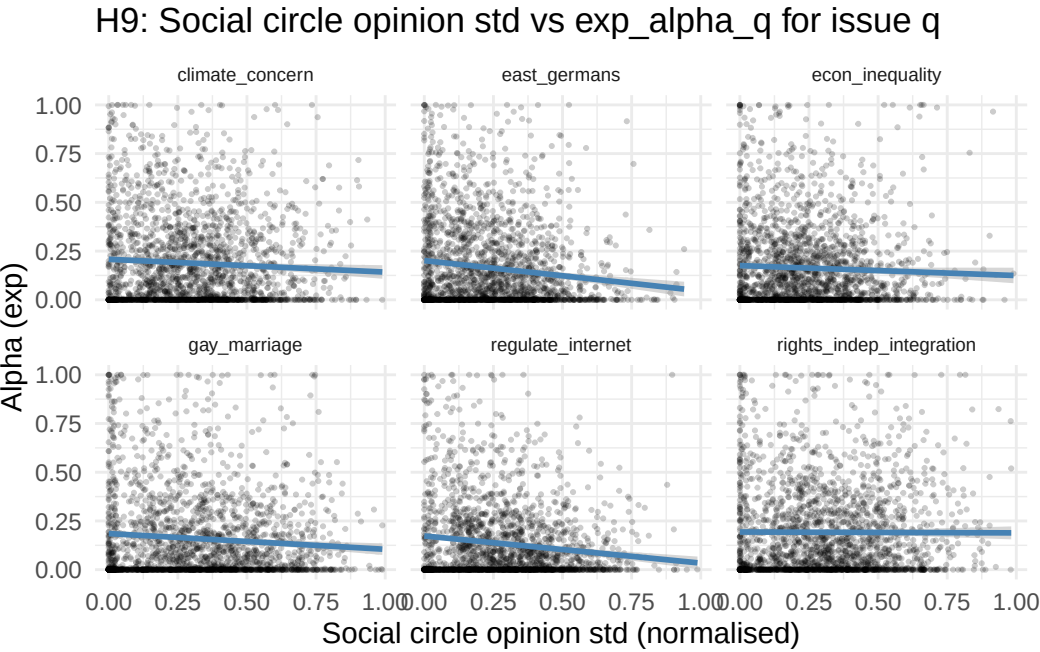
=== H9: social circle std predicts weights (exp) ===

Term	Estimate	SE	t
Intercept	0.188	0.010	19.555
Social circle std	-0.079	0.010	-8.004
Wave	0.001	0.004	0.345

p (one-sided, H1: beta < 0) = 0.0000
Direction correct (beta < 0): Yes | Reject H0 at alpha = 0.05: Yes

```
ggplot(df_h9, aes(x = var_sc, y = alpha)) +  
  geom_point(alpha = 0.2, size = 0.4) +  
  geom_smooth(method = "lm", se = TRUE, color = "steelblue") +  
  facet_wrap(~ issue, ncol = 3) +  
  labs(title = paste0("H9: Social circle opinion ",variance_type," vs ",k,"_alpha_q for issue q"),  
       x = paste0("Social circle opinion ",variance_type," (normalised)",  
       y = paste0("Alpha (", k, ")")) +  
  theme_minimal() +  
  theme(strip.text = element_text(size = 7))
```

`geom\_smooth()` using formula = 'y ~ x'



```
ggsave(paste0("h9_plot_",k,"_",variance_type,".png"), width = 16/2.54, height = 9/2.54)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
variance_type = "std"
k = "corrS"
```

```
models_h9 <- setNames(questions_sc, questions_sc) |>
  lapply(function(q) {
    df_q <- df_p |>
      filter(wave %in% c(1, 2)) |>
      select(
        id, wave,
        var_sc = all_of(paste0(variance_type, "_socialCircle_ops_", q)),
        alpha = all_of(paste0(k, "_alpha_", q))
      ) |>
      filter(!is.na(alpha), !is.na(var_sc))

    lmer(alpha ~ var_sc + wave + (1 | id), data = df_q)
  })
```

```
cat(sprintf("=== H9-separate: social circle %s predicts weights (%s) for each issue ===\n", variance_type, k) )
results_h9 <- purrr::map_dfr(questions_sc, function(q) {
  cr <- summary(models_h9[[q]])$coefficients["var_sc", ]
  t_stat = cr["t value"]
  p1 <- if (t_stat < 0) pt(t_stat, Inf) else 1 - pt(t_stat, Inf)
  data.frame(
    Issue      = q,
    beta       = round(cr["Estimate"], 4),
    SE         = round(cr["Std. Error"], 4),
    t          = round(cr["t value"], 4),
    p_one      = round(p1, 4),
    correct    = cr["Estimate"] < 0,
    reject_05  = cr["Estimate"] < 0 & p1 < 0.05
  )
})

print(kable(results_h9, format = "simple"))
```

=== H9-separate: social circle std predicts weights (corrS) for each issue ===

	Issue	beta	SE	t	p_one	correct	reject_05
Estimate...1	climate_concern	-0.0647	0.0276	-2.3433	0.0096	TRUE	TRUE
Estimate...2	gay_marriage	-0.1176	0.0246	-4.7780	0.0000	TRUE	TRUE
Estimate...3	rights_indep_integration	-0.0546	0.0266	-2.0560	0.0199	TRUE	TRUE
Estimate...4	econ_inequality	-0.1530	0.0295	-5.1896	0.0000	TRUE	TRUE
Estimate...5	regulate_internet	-0.1965	0.0267	-7.3532	0.0000	TRUE	TRUE
Estimate...6	east_germans	-0.2255	0.0271	-8.3176	0.0000	TRUE	TRUE

3.3.2 H10

3.3.2.1 H10. Social circle variance predicts reported issue importance.

We model reported issue importance of participant n as

$$w_{n,q} = \beta_0 + \beta_q \cdot \text{var}_{n,q} + u_q + v_n + \epsilon_{n,q} . \tag{11}$$

where v_n is the average importance of all issues assigned by participant n .

The prediction is $\beta_q < 0$. Issues are rated as more important when social circle variance is smaller.

```
variance_type <- "std"

df_h10 <- df_p %>%
  filter(wave %in% c(1,2)) %>%
  pivot_longer(
```

```
cols = c(all_of(paste0("w_", questions_sc)), paste0(variance_type, "_socialCircle_ops_", questions_sc)),
names_to = c(".value", "issue"),
names_pattern = all_of(paste0("^(", variance_type, "_socialCircle_ops|w)_(.*)"))
)%>%
rename(var_sc = all_of(paste0(variance_type, "_socialCircle_ops")))
```

Warning: Using `all\_of()` outside of a selecting function was deprecated in tidyselect 1.2.0.

i See details at

[<https://tidyselect.r-lib.org/reference/faq-selection-context.html>](https://tidyselect.r-lib.org/reference/faq-selection-context.html)

```
#%>%
# group_by(id) %>%
# mutate(delta_w = w - mean(w)) %>%
# ungroup()
```

```
model_h10 <- lmer(w ~ var_sc + wave + (1 | issue) + (1 | id) , data = df_h10)
```

```
cat(sprintf("=== H10: social circle %s predicts issue importance w ===\n", variance_type) )
```

```
coef_row <- summary(model_h10)$coefficients["var_sc", ]
df_resid <- coef_row["df"]
t_stat <- coef_row["t value"]
df_resid <- nrow(df_h10) - length(fixef(model_h10)) # approximate
p_one <- if (t_stat < 0) pt(t_stat, df_resid) else 1 - pt(t_stat, df_resid)
cat()
cat(sprintf("beta = %.4f\n", coef_row["Estimate"] ),
sprintf("SE = %.4f\n", coef_row["Std. Error"] ),
sprintf("t = %.4f\n", t_stat),
sprintf("p (one-sided, H1: beta < 0) = %.4f\n", p_one),
sprintf("Direction correct (beta < 0): %s\n", coef_row["Estimate"] < 0),
sprintf("Reject H0 at alpha=0.05: %s\n", coef_row["Estimate"] < 0 & p_one < 0.05))
```

```
=== H10: social circle std predicts issue importance w ===
```

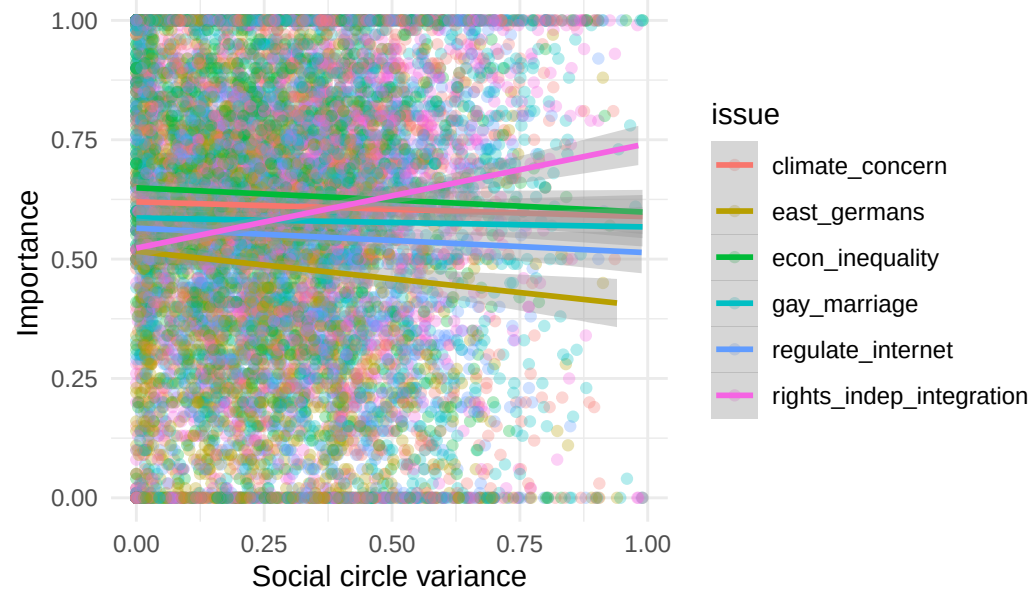
```
beta = 0.0053
SE = 0.0141
t = 0.3748
p (one-sided, H1: beta < 0) = 0.3539
Direction correct (beta < 0): FALSE
Reject H0 at alpha=0.05: FALSE
```

Plot by issue

```
df_h10 %>%
ggplot(aes(x = var_sc, y = w, color=issue)) +
geom_point(alpha = 0.3) +
geom_smooth(method = "lm", se = TRUE) +
labs(x = "Social circle variance",
y = "Importance ",
title = sprintf("H10: Social circle %s vs. relative importance", variance_type)) +
theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'

H10: Social circle std vs. relative importance



Separate Model with different slopes per issue

```
cat(sprintf("=== H10-separate: social circle %s predicts issue importance w for each issue ===\n", variance_type) )

for (q in questions_sc) {

  df_q <- df_h10 %>%
    filter(issue==q) %>%
    select(id, wave, issue, w, var_sc)

  m <- lmer(w ~ var_sc + (1 | id) + wave , data = df_q)

  s <- summary(m)
  beta <- s$coefficients["var_sc", "Estimate"]
  p_val <- s$coefficients["var_sc", "Pr(>|t|)"]

  cat(sprintf("%s (beta=%.2f --> p = %.3f | Reject H0: %s)\n",
    q, beta, p_val, (p_val < 0.05) & (beta < 0)))
}
```

```
=== H10-separate: social circle std predicts issue importance w for each issue ===
climate_concern (beta=-0.01 --> p = 0.795 | Reject H0: FALSE)
gay_marriage (beta=-0.02 --> p = 0.556 | Reject H0: FALSE)
rights_indep_integration (beta=0.18 --> p = 0.000 | Reject H0: FALSE)
econ_inequality (beta=-0.06 --> p = 0.055 | Reject H0: FALSE)
regulate_internet (beta=-0.05 --> p = 0.088 | Reject H0: FALSE)
east_germans (beta=-0.11 --> p = 0.002 | Reject H0: TRUE)
```

3.3.3 H11

3.3.3.1 H11. Overall social circle variance helps predict likeability rating of typical voters.

In H2 (equation 2), we modelled the participant n 's likeability rating of the typical voter of political party r as

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + u_n + \epsilon_{n,r}. \quad (12)$$

Here, we extend this model as

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + \beta_2 \cdot \sum_q \text{var}_{n,q} + v_n + \epsilon_{n,r} \quad (13)$$

where the new β_2 -term represents the influence of overall social circle belief variation on likeability.

The prediction is $\beta_2 > 0$ and that equation 14 improves the predictions of the baseline model from **H2** (equation 2/12) using R2, AIC, and BIC as model quality indicators. Voters are more liked in general if the participant's overall social circle variance is higher.

{TODO: This hypothesis is kind of wrong. We have 1 avg_social_circle_std per person and 7 pixel_dist and sympathy for 7 parties. But the avg_social_circle_std is already included in the random intercept.

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + \beta_2 \cdot \sum_q \text{var}_{n,q} \text{REMOVED} (+v_n) + \epsilon_{n,r} \quad (14)$$

```

}

variance_type = "std"

df_h2And11 <- df_diff %>%
  filter(dot1 == "self", dot2 %in% partiesVars, wave == 2) %>%
  select(id, sympathy, party, pixel_dist)

avg_var_sc <- df_p %>%
  filter(wave == 2) %>%
  select(id, all_of(paste0(variance_type, "_socialCircle_ops_", questions_sc))) %>%
  drop_na() %>%
  mutate(avg_var_sc = rowMeans(across(all_of(paste0(variance_type, "_socialCircle_ops_", questions_sc))))) %>%
  select(id, avg_var_sc)

df_h11 <- df_h2And11 %>%
  left_join(avg_var_sc, by = "id") %>%
  mutate(
    pixel_dist_z = as.numeric(scale(pixel_dist)),
    avg_var_sc_z = as.numeric(scale(avg_var_sc))
  )

df_h11_agg <- df_h11 %>%
  group_by(id, avg_var_sc_z) %>%
  summarise(mean_sympathy = mean(sympathy, na.rm = TRUE), .groups = "drop")

model_h11_agg <- lm(mean_sympathy ~ avg_var_sc_z, data = df_h11_agg)
summary(model_h11_agg)

```

Call:

```
lm(formula = mean_sympathy ~ avg_var_sc_z, data = df_h11_agg)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.41923	-0.07643	0.00217	0.07832	0.58223

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.420409	0.004284	98.145	<2e-16 ***
avg_var_sc_z	0.007005	0.004284	1.635	0.102

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1327 on 958 degrees of freedom
(1 observation deleted due to missingness)
Multiple R-squared: 0.002783, Adjusted R-squared: 0.001742
F-statistic: 2.674 on 1 and 958 DF, p-value: 0.1023

Now let's go to treating each party as separate data point (for one participant)

```

model_ext <- lmer(
  sympathy ~ pixel_dist_z * avg_var_sc_z + (1 + pixel_dist_z | id),
  data = df_h11, REML = FALSE
)

summary(model_ext)

# Baseline (H2): y_{n,r} = 0 + 1*d_{n,self-r} + u_n +
model_base <- lmer(sympathy ~ pixel_dist_z + (1 | id),
  data = df_h11, REML = FALSE)

# Extended (H11): y_{n,r} = 0 + 1*d_{n,self-r} + 2*Σvar_{n,q} + v_n +
model_ext <- lmer(sympathy ~ pixel_dist_z + avg_var_sc_z + (1 | id),
  data = df_h11, REML = FALSE)

```

```
# Corrected df for between-person predictor (level-2: n_participants - 2)
n_participants <- n_distinct(df_h11$id)
df_corrected    <- n_participants - 2

beta2 <- fixef(model_ext)["avg_var_sc_z"]
beta1 <- fixef(model_ext)["pixel_dist_z"]
se_b2 <- sqrt(diag(vcov(model_ext))["avg_var_sc_z"])
se_b1 <- sqrt(diag(vcov(model_ext))["pixel_dist_z"])
t_stat <- beta2 / se_b2
p_two <- 2 * pt(abs(t_stat), df = df_corrected, lower.tail = FALSE)
p_one <- if (t_stat > 0) p_two / 2 else 1 - p_two / 2

r2_base <- r2(model_base)
r2_ext  <- r2(model_ext)

# Named access to marginal (R2m) and conditional (R2c) R²
r2m_base <- r2_base$R2_marginal
r2c_base <- r2_base$R2_conditional
r2m_ext  <- r2_ext$R2_marginal
r2c_ext  <- r2_ext$R2_conditional

cat(
  sprintf("H11 Test:\n"),
  sprintf("  1 (pixel_dist standardised)          = %.4f\n", beta1),
  sprintf("    (social circle variance standardised)    = %.4f\n", beta2),
  sprintf("  SE                                = %.4f\n", se_b2),
  sprintf("  t                                  = %.4f\n", t_stat),
  sprintf("  df (corrected)                        = %d\n", df_corrected),
  sprintf("  p two-sided                          = %.6f\n", p_two),
  sprintf("  p one-sided ( > 0)                   = %.6f\n", p_one),
  sprintf("  Reject H0 ( 0)                      = %s\n", p_one < 0.05 && beta2 > 0),
  sprintf("  R² marginal baseline / extended = %.4f / %.4f (Delta = %+.4f)\n",
    r2m_base, r2m_ext, r2m_ext - r2m_base),
  sprintf("  R² conditional baseline / extended = %.4f / %.4f (Delta = %+.4f)\n",
    r2c_base, r2c_ext, r2c_ext - r2c_base),
  sprintf("  AIC baseline / extended = %.2f / %.2f (Delta = %+.2f)\n",
    AIC(model_base), AIC(model_ext), AIC(model_ext) - AIC(model_base)),
  sprintf("  BIC baseline / extended = %.2f / %.2f (Delta = %+.2f)\n",
    BIC(model_base), BIC(model_ext), BIC(model_ext) - BIC(model_base))
)
```

Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's method [lmerModLmerTest]
Formula: sympathy ~ pixel_dist_z * avg_var_sc_z + (1 + pixel_dist_z | id)
Data: df_h11

AIC	BIC	logLik	-2*log(L)	df.resid
-24.7	29.8	20.3	-40.7	6712

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.3174	-0.5732	0.0221	0.6024	4.0515

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
id	(Intercept)	0.020588	0.14349	
	pixel_dist_z	0.008848	0.09407	0.61
Residual		0.044355	0.21061	

Number of obs: 6720, groups: id, 960

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	0.400433	0.005465	784.293772	73.277	< 2e-16 ***
pixel_dist_z	-0.247590	0.004622	678.060737	-53.567	< 2e-16 ***
avg_var_sc_z	-0.017930	0.005477	791.887019	-3.273	0.00111 **

```
pixel_dist_z:avg_var_sc_z  -0.022370    0.004631 684.651175 -4.831 1.68e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Correlation of Fixed Effects:

```
(Intr) pxl_d_avg___
pixel_dst_z 0.417
avg_vr_sc_z 0.000 0.035
pxl_ds_:___ 0.035 0.022 0.404
```

H11 Test:

```
1 (pixel_dist standardised)      = -0.2261
(social circle variance standardised) = -0.0135
SE                                = 0.0052
t                                = -2.5813
df (corrected)                   = 959
p two-sided                      = 0.009992
p one-sided ( > 0)               = 0.995004
Reject H0 ( 0)                  = FALSE
R² marginal baseline / extended = 0.4208 / 0.4211 (Delta = +0.0002)
R² conditional baseline / extended = 0.5802 / 0.5775 (Delta = -0.0027)
AIC baseline / extended = 313.82 / 303.83 (Delta = -9.99)
BIC baseline / extended = 341.08 / 337.89 (Delta = -3.19)
```

3.3.4 H12a

3.3.4.1 H12a. Social circle variance predicts perceived political polarisation on a given issue.

We model perceived polarisation on issue q as

$$P_{n,q} = \beta_0 + \beta_q \cdot var_{n,q} + u_n + \epsilon_{n,q}. \tag{15}$$

The prediction is $\beta_q < 0$. Participants with less variation in their social circles with respect to issue q perceive higher polarisation on that issue. %We test this via a single F-test on the joint significance of $\{\beta_q\}$, with significance threshold $\alpha = 0.05$.

```
variance_type = "var"
df_h12a <- df_p %>%
  filter(wave %in% c(1,2)) %>%
  select(-"P_tot") %>%
  pivot_longer(
    cols = c(all_of(paste0("P_", questions_sc)), all_of(paste0(variance_type, "_socialCircle_ops_", questions_sc))),
    names_to = c(".value", "issue"),
    names_pattern = all_of(paste0("^(",variance_type,"_socialCircle_ops|P)_(.*)"))
  ) %>%
  rename(var_sc = paste0(variance_type,"_socialCircle_ops"))

model_h12a <- lmer(P ~ var_sc + (1 | issue) + (1 | id) + wave , data = df_h12a)
summary(model_h12a)

# One-sided p-value for beta < 0
coef_row <- summary(model_h12a)$coefficients["var_sc", ]
t_stat <- coef_row["t value"]
df_resid <- nrow(df_h12a) - length(fixef(model_h12a)) # approximate
p_one <- if (t_stat < 0) pt(t_stat, df_resid) else 1 - pt(t_stat, df_resid)

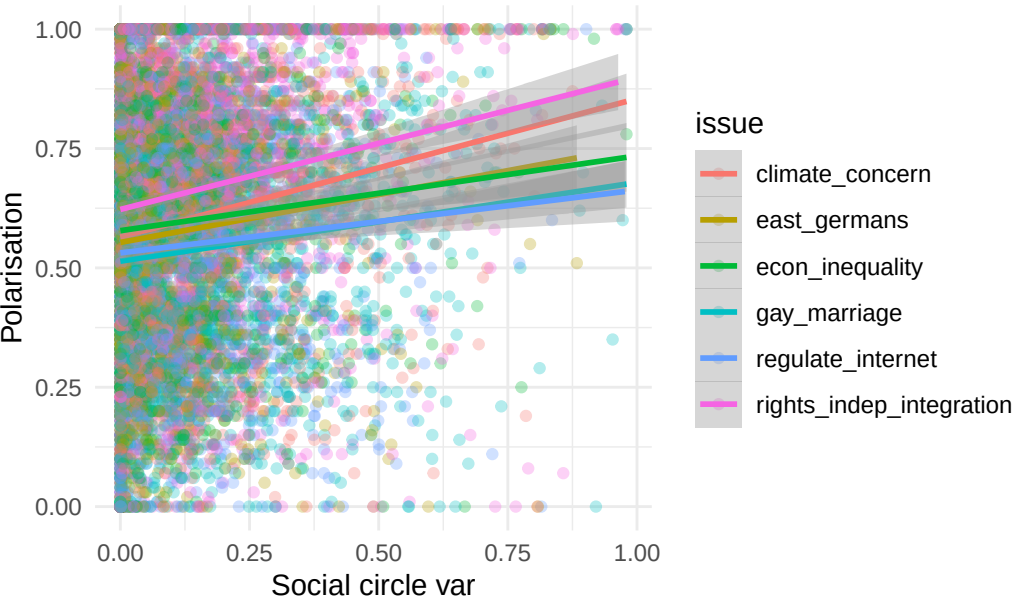
cat(sprintf("beta = %.4f\n", coef_row["Estimate"]),
  sprintf("SE = %.4f\n", coef_row["Std. Error"]),
  sprintf("t = %.4f\n", t_stat),
  sprintf("p (one-sided, H1: beta < 0) = %.4f\n", p_one),
  sprintf("Direction correct (beta < 0): %s\n", coef_row["Estimate"] < 0),
  sprintf("Reject H0 at alpha=0.05: %s\n", coef_row["Estimate"] < 0 & p_one < 0.05))

df_h12a %>%
  ggplot(aes(x = var_sc, y = P, color=issue)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = paste0("Social circle ",variance_type),
```

```
y = "Polarisation",
title = paste0("H12a: Social circle ",variance_type," vs. issue polarisation")) +
theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

H12a: Social circle var vs. issue polarisation



Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]
Formula: P ~ var_sc + (1 | issue) + (1 | id) + wave
Data: df_h12a

REML criterion at convergence: -2256.3

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.9871	-0.6042	0.0943	0.6562	3.3072

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.014176	0.11906
issue	(Intercept)	0.001947	0.04412
Residual		0.042435	0.20600

Number of obs: 12774, groups: id, 1227; issue, 6

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	5.654e-01	1.859e-02	5.558e+00	30.415	2.14e-07 ***
var_sc	2.195e-01	1.593e-02	1.216e+04	13.777	< 2e-16 ***
wave2	-1.105e-02	3.861e-03	1.241e+04	-2.861	0.00423 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr) var_sc
var_sc	-0.098
wave2	-0.084 -0.046

beta = 0.2195
SE = 0.0159
t = 13.7769
p (one-sided, H1: beta < 0) = 0.0000
Direction correct (beta < 0): FALSE
Reject H0 at alpha=0.05: FALSE

3.3.5 H12b

3.3.5.1 H12b. Social circle variance predicts perceived political polarisation.

We model overall perceived polarisation as

$$P_{n,tot} = \beta_0 + \beta_1 \cdot \sum_q var_{n,q} + \epsilon_n. \tag{16}$$

The prediction is \$ _1 < 0\$. Participants with less variation in their social circles averaged over all issues perceive more political polarisation.
%We test this via a t-test with significance threshold $\alpha = 0.05$.

```
df_h12b <- df_p %>%
  select(id, wave, all_of(paste0(variance_type,"_socialCircle_ops_", questions_sc)), P_tot) %>%
  drop_na() %>%
  mutate(avg_var_sc = rowSums(across(all_of(paste0(variance_type,"_socialCircle_ops_", questions_sc)))))

model <- lmer(P_tot ~ avg_var_sc + (1 | id),
  data = df_h12b, REML = TRUE)

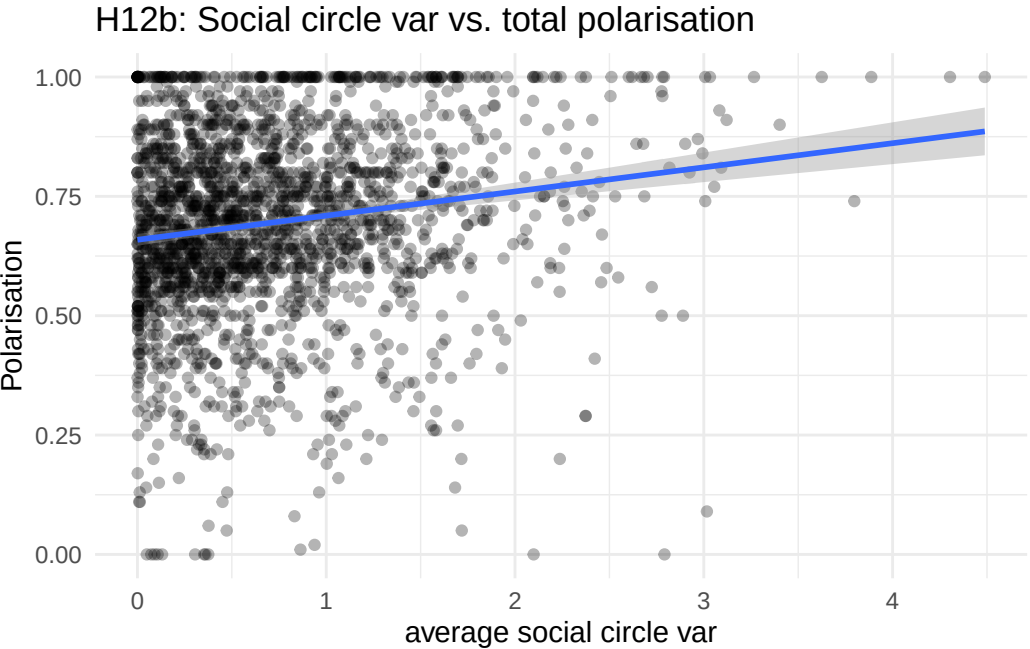
n_participants <- n_distinct(df_h12b$id)
df_corrected <- n_participants - 2

beta <- fixef(model)["avg_var_sc"]
se_b <- sqrt(diag(vcov(model)))["avg_var_sc"]
t_stat <- beta / se_b
p_two <- 2 * pt(abs(t_stat), df = df_corrected, lower.tail = FALSE)
p_one <- p_two / 2

cat(sprintf("H12b Test (one-sided, < 0):\n"),
  sprintf(" = %.4f\n", beta),
  sprintf(" t = %.4f\n", t_stat),
  sprintf(" p (two-sided) = %.4f\n", p_two),
  sprintf(" p (one-sided) = %.4f\n", p_one),
  sprintf(" Reject H0 ( < 0): %s\n", p_one < 0.05 && beta < 0))

df_h12b %>%
  ggplot(aes(x = avg_var_sc, y = P_tot)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = paste0("average social circle ",variance_type),
    y = "Polarisation",
    title = paste0("H12b: Social circle ",variance_type," vs. total polarisation")) +
  theme_minimal()
```

``geom_smooth()`` using formula = 'y ~ x'



```
H12b Test (one-sided, < 0):
  = 0.0418
t   = 5.8658
p   (two-sided) = 0.0000
p   (one-sided) = 0.0000
Reject H0 (< 0): FALSE
```

3.3.6 H13

3.3.6.1 H13. Reported issue importance predicts perceived distance.

In equation ??, we fix the issue weights to $\alpha_{n,q} = w_{n,q}$. Then, we model the perceived distance between two individuals i and j (using the exponential kernel; see equations ?? and ??) as

$$d_{n,ij} = 1 - \exp \left(-\hat{b}_n \cdot \left(\sum_q w_{n,q} \cdot |x_{n,i,q} - x_{n,j,q}| \right)^{\hat{c}_n} \right) \tag{17}$$

with fit parameters $\hat{b}_n$, and $\hat{c}_n$.

The prediction is that this model (with $\hat{\alpha}_{n,q} = w_{n,q}$) outperforms (i) an unweighted model with $\hat{\alpha}_{n,q} = 1/6$ and (ii) a population weight model with $\hat{\alpha}_{n,q} = 1/n \cdot \sum_n w_{n,q}$. We use R2, AIC, and BIC as model quality indicators.

```
# weighted_l1 <- function(deltas, alpha) {
#   as.vector(as.matrix(deltas) %*% as.numeric(alpha))
# }

linear_kernel <- function(x, a, b) a + b * x

exp_kernel    <- function(x, a, b) 1 - exp(-a * (x ^ b))

weighted_mean_delta <- function(d, w) {
  w_sum <- rowSums(w, na.rm = TRUE)
  num   <- rowSums(d * w, na.rm = TRUE)
  ifelse(w_sum == 0, 0, num / w_sum)
}

r2_vec <- function(y, pred_mat) {
  ss_tot <- sum((y - mean(y, na.rm = TRUE))^2, na.rm = TRUE)
  1 - colSums((pred_mat - y)^2, na.rm = TRUE) / ss_tot
}

compute_aic_bic <- function(mse, k, n) {
  rss <- mse * n
  tibble(
    aic = n * log(rss / n) + 2 * k,
    bic = n * log(rss / n) + k * log(n)
  )
}

col_deltas    <- paste0("deltaX_", questions_sc)
col_alpha_lin <- paste0("linear_alpha_", questions_sc)
col_alpha_exp <- paste0("exp_alpha_",   questions_sc)
col_weights   <- paste0("w_",          questions_sc)

alphas_population_lin <- df_p |> select(all_of(col_alpha_lin)) |> colMeans()
alphas_population_exp <- df_p |> select(all_of(col_alpha_exp)) |> colMeans()

uniform_alpha <- rep(1 / length(questions_sc), length(questions_sc))

df_h13_base <- df_diff |>
  filter(wave %in% c(1, 2)) |>
  left_join(
    df_p |> select(id, wave, all_of(col_weights), all_of(col_alpha_lin), all_of(col_alpha_exp)),
    by = c("id", "wave"),
    suffix = c("", ".drop") # keep df_diff version, mark df_p duplicates
  ) |>
```

```

select(-ends_with(".drop")) # remove the conflicting df_p columns

# -- Extract matrices outside of mutate -----
.d      <- as.matrix(df_h13_base[, col_deltas])
.a_uniform <- matrix(1 / length(questions_sc), nrow = nrow(df_h13_base), ncol = length(questions_sc))
.a_pop_exp <- matrix(alphas_population_exp,    nrow = nrow(df_h13_base), ncol = length(questions_sc), byrow = TRUE)
.a_pop_lin <- matrix(alphas_population_lin,    nrow = nrow(df_h13_base), ncol = length(questions_sc), byrow = TRUE)
.a_fit_exp <- as.matrix(df_h13_base[, col_alpha_exp])
.a_fit_lin <- as.matrix(df_h13_base[, col_alpha_lin])
.a_weights <- {
  w      <- as.matrix(df_h13_base[, col_weights])
  w_sum <- rowSums(w, na.rm = TRUE)
  ifelse(w_sum == 0, 0, w / w_sum)
}

# -- Weighted sums -----
.x_uniform <- rowSums(.d * .a_uniform)
.x_pop_exp <- rowSums(.d * .a_pop_exp)
.x_pop_lin <- rowSums(.d * .a_pop_lin)
.x_fit_exp <- rowSums(.d * .a_fit_exp)
.x_fit_lin <- rowSums(.d * .a_fit_lin)
.x_weights <- rowSums(.d * .a_weights)

# -- Attach predictions -----
df_h13 <- df_h13_base |>
  mutate(
    predicted_d_uniform_exp = exp_kernel(.x_uniform, exp_param1, exp_param2),
    predicted_d_uniform_lin = linear_kernel(.x_uniform, linear_param1, linear_param2),
    predicted_d_population_exp = exp_kernel(.x_pop_exp, exp_param1, exp_param2),
    predicted_d_population_lin = linear_kernel(.x_pop_lin, linear_param1, linear_param2),
    predicted_d_fit_exp = exp_kernel(.x_fit_exp, exp_param1, exp_param2),
    predicted_d_fit_linear = linear_kernel(.x_fit_lin, linear_param1, linear_param2),
    predicted_d_weights = exp_kernel(.x_weights, exp_param1, exp_param2)
  )

pred_cols <- c(
  "predicted_d_weights",
  "predicted_d_population_exp", "predicted_d_population_lin",
  "predicted_d_uniform_exp",    "predicted_d_uniform_lin",
  "predicted_d_fit_linear",     "predicted_d_fit_exp"
)

err_cols <- str_replace(pred_cols, "predicted_d_", "err_")

df_h13 <- df_h13 |>
  mutate(across(
    all_of(pred_cols),
    ~ pixel_dist - .x,
    .names = "err_{.col}"
  ))

df_h13 <- df_h13 |>
  rename_with(~ str_replace(.x, "err_predicted_d_", "err_"), starts_with("err_"))

# -- Model comparison table -----

y      <- df_h13$pixel_dist
n_obs  <- sum(!is.na(y))
n_pw   <- df_h13 |> distinct(id, wave) |> nrow() # participant-waves
n_q    <- length(questions_sc)

pred_mat <- df_h13 |> select(all_of(pred_cols)) |> as.matrix()
colnames(pred_mat) <- str_remove(pred_cols, "predicted_d_")

# k = number of free parameters per model

```

```

# Kernel always contributes 2 (a, b); alphas only when individually fitted
k_lookup <- tribble(
  ~model,      ~k,
  "weights",    2,                # kernel only; weights come from data
  "population_exp", 2 + n_q,        # kernel + population alphas
  "population_lin", 2 + n_q,
  "uniform_exp",  2,                # kernel only; no alpha parameters
  "uniform_lin",  2,
  "fit_exp",      (2 + n_q) * n_pw, # kernel + per-participant-wave alphas
  "fit_linear",   (2 + n_q) * n_pw
)

baseline_err <- (pred_mat[, "uniform_exp"] - y)^2

comparison_tbl <- tibble(
  model = colnames(pred_mat),
  mse   = colMeans((pred_mat - y)^2, na.rm = TRUE),
  rmse  = sqrt(mse),
  r2     = r2_vec(y, pred_mat)
) |>
mutate(
  rel_improve = (mse[model == "uniform_exp"] - mse) / mse[model == "uniform_exp"],
  # Wilcoxon vs uniform_exp baseline (no duplicate row needed)
  p_wilcox = map_dbl(model, \(m) {
    if (m == "uniform_exp") return(NA_real_)
    wilcox.test(
      (pred_mat[, m] - y)^2,
      baseline_err,
      paired = TRUE
    )$p.value
  }),
  mean_diff_vs_uniform = map_dbl(model, \(m) {
    if (m == "uniform_exp") return(NA_real_)
    mean((pred_mat[, m] - y)^2 - baseline_err, na.rm = TRUE)
  })
) |>
left_join(k_lookup, by = "model") |>
mutate(
  aic_bic = map2(mse, k, ~ compute_aic_bic(.x, .y, n_obs)),
  aic      = map_dbl(aic_bic, "aic"),
  bic      = map_dbl(aic_bic, "bic")
) |>
select(-aic_bic, -k) |>
arrange(mse)

print(comparison_tbl)

```

```

# A tibble: 7 x 9
  model    mse  rmse      r2 rel_improve p_wilcox mean_diff_vs_uniform    aic
<chr> <dbl> <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
1 fit_~ 0.0162 0.127 0.509    0.237    0      -0.00505 -1.10e6
2 fit_~ 0.0164 0.128 0.505    0.230    0      -0.00490 -1.10e6
3 popu~ 0.0212 0.146 0.358    0.00238 2.54e-71 -0.0000506 -1.06e6
4 unif~ 0.0213 0.146 0.357    0        NA      NA        -1.06e6
5 weig~ 0.0330 0.182 0.00175 -0.552    0      0.0117   -9.41e5
6 unif~ 0.0431 0.208 -0.301   -1.02    4.13e-14 0.0218   -8.68e5
7 popu~ 0.0466 0.216 -0.407   -1.19    1.29e- 2 0.0253   -8.46e5
# i 1 more variable: bic <dbl>

```

3.3.6.2 Analysis

```

# 1. Per-participant performance (main result)
per_person_r2 <- df_h13 |>
  group_by(id) |>
  summarise(across(

```

```

all_of(pred_cols),
~ {
  valid <- !is.na(.x) & !is.na(pixel_dist)
  if (sum(valid) < 3) return(NA_real_)

  ss_res <- sum((pixel_dist[valid] - .x[valid])^2)
  ss_tot <- sum((pixel_dist[valid] - mean(pixel_dist[valid]))^2)

  if (ss_tot == 0) return(NA_real_)

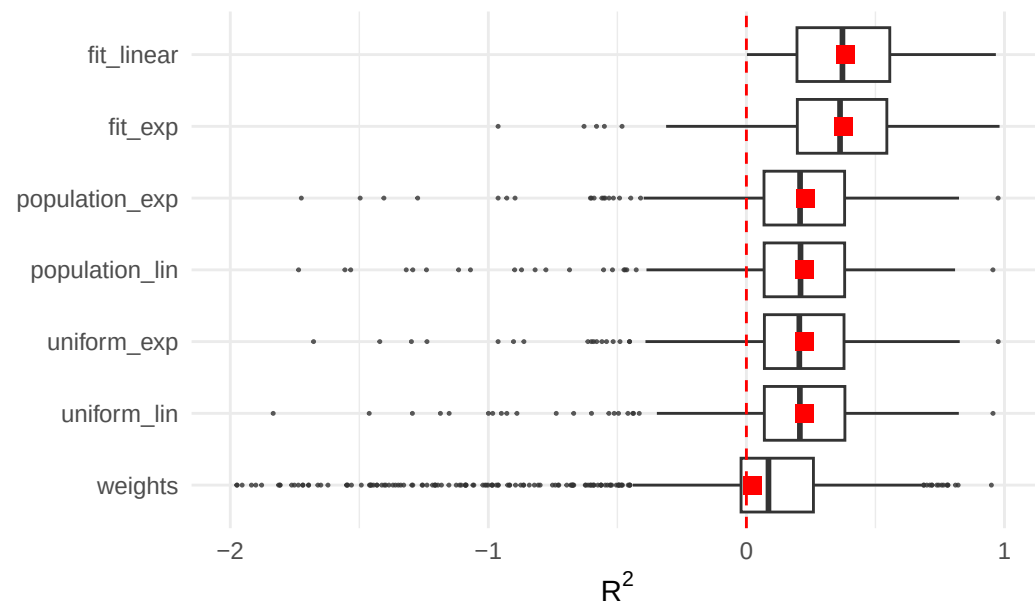
  1 - ss_res / ss_tot
},
.names = "r2_{.col}"
)) |>
pivot_longer(
  starts_with("r2_"),
  names_to = "model",
  values_to = "r2"
) |>
mutate(model = str_remove(model, "r2_predicted_d_"))
ggplot(
  per_person_r2,
  aes(
    x = reorder(model, r2, median, na.rm = TRUE),
    y = r2
  )
) +
geom_boxplot(outlier.alpha = .8, outlier.size=0.2) +
stat_summary(
  fun = mean,
  geom = "point",
  shape = 15,      # filled square
  size = 3,
  colour = "red"
) +
geom_hline(
  yintercept = 0,
  colour = "red",
  linetype = "dashed"
) +
coord_flip() +
labs(
  title = "Per-participant R2",
  x = NULL,
  y = expression(R2)
) +
ylim(-2,1)+
theme_minimal()

```

Warning: Removed 94 rows containing non-finite outside the scale range (``stat_boxplot()``).

Warning: Removed 94 rows containing non-finite outside the scale range (``stat_summary()``).

Per-participant R^2

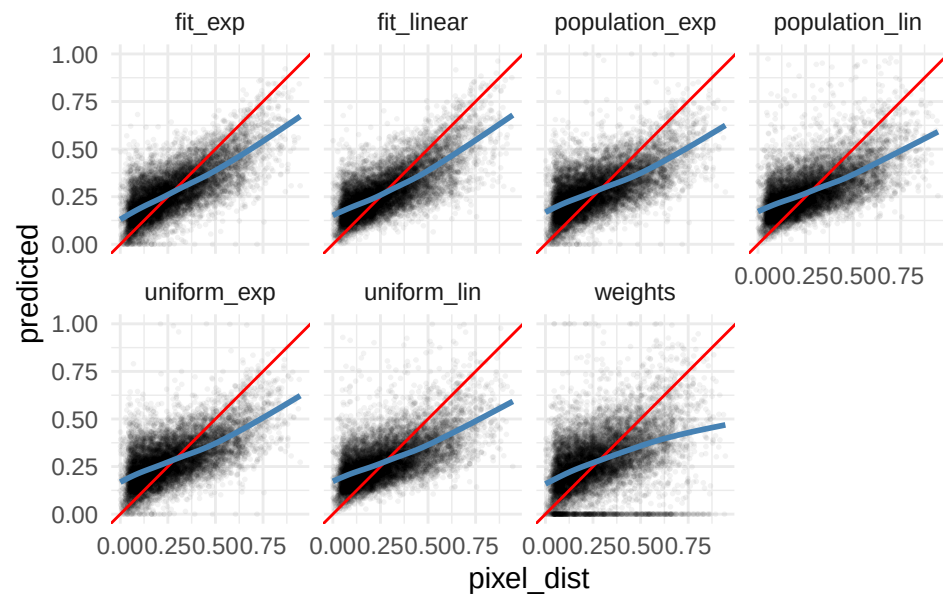


```
# 2. Predicted vs observed
df_plot <- df_h13 |>
  slice_sample(n = 10000)
pred_long <- df_plot |>
  select(pixel_dist, all_of(pred_cols)) |>
  pivot_longer(
    -pixel_dist,
    names_to = "model",
    values_to = "predicted"
  ) |>
  mutate(model = str_remove(model, "predicted_d_"))
ggplot(
  pred_long,
  aes(pixel_dist, predicted)
) +
  geom_point(
    alpha = .05,
    size = .3
  ) +
  geom_abline(
    slope = 1,
    intercept = 0,
    colour = "red"
  ) +
  geom_smooth(
    method = "gam",
    formula = y ~ s(x, bs = "cs"),
    se = FALSE,
    colour = "steelblue"
  ) +
  facet_wrap(~ model, ncol=4) +
  ylim(0,1)+
  coord_equal() +
  theme_minimal() +
  labs(
    title = "Predicted vs observed"
  )
)
```

Warning: Removed 70 rows containing non-finite outside the scale range (``stat_smooth()``).

Warning: Removed 70 rows containing missing values or values outside the scale range (``geom_point()``).

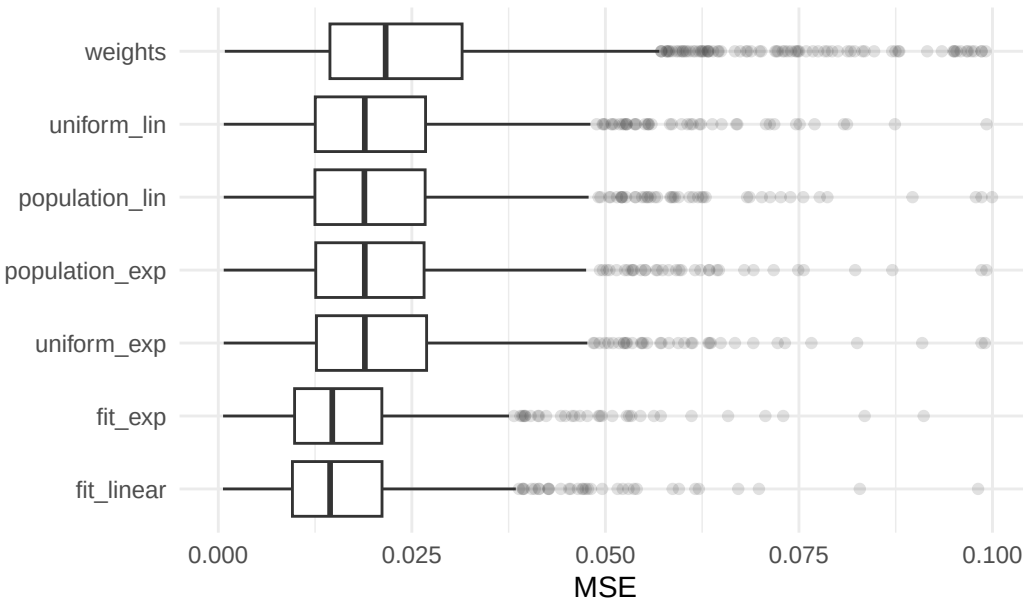
Predicted vs observed



```
# 3. Participant-level error distribution
participant_mse <- df_h13 |>
  group_by(id) |>
  summarise(across(
    all_of(err_cols),
    ~ mean(.x^2, na.rm = TRUE)
  ))
participant_mse_long <- participant_mse |>
  pivot_longer(
    -id,
    names_to = "model",
    values_to = "mse"
  ) |>
  mutate(model = str_remove(model, "^err_"))
ggplot(
  participant_mse_long,
  aes(
    x = reorder(model, mse, median),
    y = mse
  )
) +
  geom_boxplot(outlier.alpha = .15) +
  coord_flip() +
  ylim(0., 0.1) +
  theme_minimal() +
  labs(
    title = "Participant-level MSE",
    x = NULL,
    y = "MSE"
  )
)
```

Warning: Removed 115 rows containing non-finite outside the scale range (``stat_boxplot()``).

Participant-level MSE



3.3.6.3 Explorative Analyses.

In addition to the preregistered hypotheses above, we explore several ways to define the function g_n (equation ??) which describes the (participant-level) aggregation of belief differences on the six items q between individuals i and j into an overall belief distance.

In particular, we model different functional forms of g_n based on theories about who consider as social context and how this social context shapes their perception of the belief space—which we refer to as a social *lens*.

Exploration of different lens models We define three different functional types of aggregation functions g_n with increasing level of complexity. First, a **simple lens** $\mathcal{S}_n$ is a 6-dimensional vector that scales (or weighs) each entry in the belief distance vector (equivalent to the weighted metric g_n in Box 1). Second, a **compression lens** $\mathcal{C}_n$ is a 6-dimensional vector, which projects the belief difference vector onto a single dimension, e.g. *liberal* vs. *conservative*, and scales the belief distance in the direction of this dimension. Third, a **full lens** $\mathcal{F}_n$ is a 6×6 matrix that transforms the belief distance vector into a subjective new space with the same number of dimensions but with rotated and scaled axes. Writing these lenses in terms of transformation matrices yields:

$$\begin{aligned} 1 \quad \text{simplified lens} \quad g_n(|x_{n,i,q} - x_{n,j,q}|) &= \left\| \begin{pmatrix} \mathcal{S}_n & 0 \\ 0 & \mathcal{S}_n \end{pmatrix} \circ \begin{pmatrix} |x_{n,i,1} - x_{n,j,1}| \\ \vdots \\ |x_{n,i,6} - x_{n,j,6}| \end{pmatrix} \right\|_p \\ 2 \quad \text{compression lens} \quad g_n(|x_{n,i,q} - x_{n,j,q}|) &= \left\| \begin{pmatrix} -\mathcal{C}_n & - \\ 0 & \dots & 0 \\ 0 & \vdots & 0 \end{pmatrix} \circ \begin{pmatrix} |x_{n,i,1} - x_{n,j,1}| \\ \vdots \\ |x_{n,i,6} - x_{n,j,6}| \end{pmatrix} \right\|_p \\ 3 \quad \text{full lens} \quad g_n(|x_{n,i,q} - x_{n,j,q}|) &= \left\| \begin{pmatrix} \mathcal{F}_{n,11} & \dots & \mathcal{F}_{n,61} \\ \vdots & \dots & \vdots \\ \mathcal{F}_{n,61} & \dots & \mathcal{F}_{n,66} \end{pmatrix} \circ \begin{pmatrix} |x_{n,i,1} - x_{n,j,1}| \\ \vdots \\ |x_{n,i,6} - x_{n,j,6}| \end{pmatrix} \right\|_p \end{aligned}$$

where $\|\cdot\|_p$ is the p-norm. Here, we choose Minkowski norm $p = 1$ or Euclidean norm $p = 2$.

The lenses $\mathcal{S}_n$, $\mathcal{C}_n$, and $\mathcal{F}_n$ of participant n depend on the belief distribution in the participant's social contexts, $x_{n,\text{context},q}$. We use two ways to define this social contexts:

- the beliefs of the seven to ten social circle contacts of the participant: $x_{n,\text{context},q} = \{x_{n,k,q} \mid k = 1 : K_n\}$.
- the beliefs of all the other participants in the participant's in-group, where in-group can mean (i) those who affiliate with the same party, $\{x_{m,\text{self},q} \mid m \text{ with party}_m = \text{party}_n\}$, (ii) those who place themselves in the same region on the left-right scale, (iii) those who have similar demographic attributes, or (iv) combinations thereof.

Then, we determine participant n 's lenses ...

- $\mathcal{S}_n$ as the inverse of the standard deviation of the $x_{n,\text{context},q}$. $\mathcal{S}_n = \left(\frac{1}{\text{Var}(x_{n,\text{context},1})}, \dots, \frac{1}{\text{Var}(x_{n,\text{context},6})} \right)$.
- $\mathcal{C}_n$ as the first (loaded) principal components of the social context beliefs. $\mathcal{C}_n = \frac{1}{\sqrt{\lambda_{n,1}}} \nu_{n,1}$.
- $\mathcal{F}_n$ as a principal component transformation, i.e as the inverse of the loaded principal component vectors of the social context beliefs. $\mathcal{F}_n = \left(\sqrt{\lambda_{n,1}} \nu_{n,1}, \dots, \sqrt{\lambda_{n,6}} \nu_{n,6} \right)^{-1}$, where $\nu_{n,\tilde{q}}$ are principal components and $\lambda_{n,\tilde{q}}$ the corresponding eigenvalues.

We predict map distances $d_{n,ij}$ using equation ?? with different choices of lenses and different definitions of social contexts. The goal of this exploration is to investigate to what extent these (theoretical) models and assumptions about participant-level social lenses describe how people perceive political differences between individuals.

3.4 Aim 4

Aim 4: Test the effects of an intervention on social circle belief variation

In wave 2, participants are randomly put in (i) a control condition, where they receive the same instruction as in wave 1, namely to choose seven to ten social contacts that they would count to their social circle, or (ii) a treatment condition, where they receive an additional instruction to choose contacts such that they represent a diverse mix of beliefs, if possible. Aim 4 is to test the effects of this instruction.

3.4.1 H14

3.4.1.1 H14. Participants in the treatment condition report higher social circle variance.

For the participants in the treatment condition T , who receive the instruction to choose diverse contacts, we measure the increase in social circle variance on issue q as

$$\Delta \text{var}_T = \sum_q (\mathbb{E}[\text{var}_{n,q}(w2) \mid \text{condition}_n = T] - \mathbb{E}[\text{var}_{n,q}(w1) \mid \text{condition}_n = T]) \tag{18}$$

The prediction is $\sim \Delta \text{var}_T > 0$ (one-sample t-test with significance threshold $\alpha = 0.05$) and $\Delta \text{var}_T > \Delta \text{var}_{\text{Control}}$ (using independent samples t-test with $\alpha = 0.05$). Social circle variance in the treatment condition T is higher than in wave 1 and higher than in the control group.

```
variance_type="std"

df_h14 <- df_p |>
  filter(id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(
    cols = matches(paste0(variance_type,"_socialCircle_ops_")),
    names_to = c(".value", "issue"),
    names_pattern = paste0("^(",variance_type,"_socialCircle_ops)_(.*)")
  ) |>
  pivot_wider(
    id_cols = c(id, issue),
    names_from = wave,
    names_prefix = "var_sc_wave",
    values_from = paste0(variance_type,"_socialCircle_ops")
  ) |>
  mutate(
    group = case_when(
      id %in% treatment_ids ~ "treatment",
      id %in% control_ids ~ "control"
    ),
    delta = var_sc_wave2 - var_sc_wave1
  )

# -- 3. Quick sanity check -----

df_h14 |>
  group_by(group) |>
  summarise(
    mean_wave1 = mean(var_sc_wave1, na.rm = TRUE),
    std_wave1 = sd(var_sc_wave1, na.rm = TRUE),
    mean_wave2 = mean(var_sc_wave2, na.rm = TRUE),
    std_wave2 = sd(var_sc_wave2, na.rm = TRUE),
    mean_delta = mean(delta, na.rm = TRUE),
    n = n()
  ) |>
  print()
t.test(var_sc_wave1 ~ group, data = df_h14)

treatment <- df_h14 |> filter(group == "treatment")
control <- df_h14 |> filter(group == "control")

# -- 5. H14-1: Deltavar_T > 0 (one-sample t-test) -----

t_test_h1 <- t.test(treatment$delta, mu = 0, alternative = "greater")

cat(sprintf("H14-1 Test (Treatment: variance in wave2 > variance in wave1):\n"),
  sprintf("  Mean delta (wave2 - wave1) = %.4f (+- %.4f)\n",
```

```

mean(treatment$delta, na.rm = TRUE),
sd(treatment$delta, na.rm = TRUE)),
sprintf("  t = %.4f\n", t_test_h1$statistic),
sprintf("  p (one-tailed) = %.4f\n", t_test_h1$p.value),
sprintf("  H1 supported: %s\n",
t_test_h1$statistic > 0 && t_test_h1$p.value < 0.05))

```

-- 6. H14-2: Deltavar_T > Deltavar_Control (independent samples t-test) -----

```

t_test_h2 <- t.test(delta ~ group, data = df_h14,
  alternative = "less", # control - treatment < 0
  var.equal = FALSE)

cat(sprintf("H14-2 Test (Treatment delta in variance > Control delta variance):\n"),
  sprintf("  Mean delta treatment = %.4f (+- %.4f)\n",
    mean(treatment$delta, na.rm = TRUE),
    sd(treatment$delta, na.rm = TRUE)),
  sprintf("  Mean delta control   = %.4f (+- %.4f)\n",
    mean(control$delta, na.rm = TRUE),
    sd(control$delta, na.rm = TRUE)),
  sprintf("  Difference (T - C)   = %.4f\n",
    mean(treatment$delta, na.rm = TRUE) - mean(control$delta, na.rm = TRUE)),
  sprintf("  t = %.4f\n", t_test_h2$statistic),
  sprintf("  p (one-tailed) = %.4f\n", t_test_h2$p.value),
  sprintf("  H2 supported: %s\n",
    t_test_h2$statistic < 0 && t_test_h2$p.value < 0.05))

```

A tibble: 2 x 7

	group	mean_wave1	std_wave1	mean_wave2	std_wave2	mean_delta	n
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<int>
1	control	0.268	0.198	0.275	0.197	0.00638	2370
2	treatment	0.285	0.199	0.311	0.205	0.0262	3042

Welch Two Sample t-test

data: var_sc_wave1 by group

t = -3.0444, df = 5101.4, p-value = 0.002343

alternative hypothesis: true difference in means between group control and group treatment is not equal to 0

95 percent confidence interval:

-0.027240504 -0.005899909

sample estimates:

mean in group control	mean in group treatment
0.2682366	0.2848068

H14-1 Test (Treatment: variance in wave2 > variance in wave1):

Mean delta (wave2 - wave1) = 0.0262 (+- 0.1884)

t = 7.6679

p (one-tailed) = 0.0000

H1 supported: TRUE

H14-2 Test (Treatment delta in variance > Control delta variance):

Mean delta treatment = 0.0262 (+- 0.1884)

Mean delta control = 0.0064 (+- 0.1780)

Difference (T - C) = 0.0198

t = -3.9599

p (one-tailed) = 0.0000

H2 supported: TRUE

3.4.2 H15

3.4.2.1 H15. Higher relative social circle variance decreases issue weight.

We model the new issue weights, $\alpha_{n,q}(w2)$, inferred in wave 2, of the participants in the treatment condition as

$$\alpha_{n,q}(w2) = \alpha_{n,q}(w1) + \beta_1 \cdot \mathbb{I}(\text{condition}_n = T) \cdot \frac{\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)}{\sum_q \text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)} + \epsilon_{n,q} \quad (19)$$

The prediction is $\beta_1 < 0$. Higher relative social circle variance on one issue decreases the weight of that issue for the participant. %We test this via a t-test with significance threshold $\alpha = 0.05$.

{UPDATE: There should be abs signs in the denominator! Otherwise small delta Var's are going to blow up things dramatically.}

{UPDATE: This does not need to depend on the Treatment condition group alone. I analysed both now}

{OPEN QUESTION: Is the denominator: $\sum_q |\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)|$ or $\sum_q |\text{var}_{n,q}(w1)|$?? I chose the first now because this is what we preregistered.}

```
variance_type = "std"
k = "exp"

df_h15 <- df_p |>
  # filter(id %in% treatment_ids) |>
  pivot_longer(
    cols = matches(paste0(variance_type, "_socialCircle_ops_")),
    names_to = c(".value", "issue"),
    names_pattern = paste0("^(", variance_type, "_socialCircle_ops_)(.*)")
  ) |>
  rename(
    var_sc = paste0(variance_type, "_socialCircle_ops")
  ) |>
  pivot_wider(
    id_cols = c(id, issue),
    names_from = wave,
    names_prefix = "w",
    values_from = var_sc
  ) |>
  rename(var_w1 = w1, var_w2 = w2) |>
  mutate(delta_var = var_w2 - var_w1)
df_h15 <- df_h15 |>
  group_by(id) |>
  mutate(
    sum_delta_var = sum(abs(delta_var), na.rm = TRUE),
    sum_w1_var = sum(abs(var_w1), na.rm = TRUE),
    rel_delta_var = if_else(sum_w1_var > 0.05,
                           delta_var / sum_delta_var,
                           NA_real_)
  ) |>
  ungroup()

df_weights <- df_p |>
  # filter(id %in% treatment_ids) |>
  pivot_longer(
    cols = matches(paste0("^", k, "_alpha_")),
    names_to = c(".value", "issue"),
    names_pattern = paste0("^(", k, "_alpha_)(.*)")
  ) |>
  pivot_wider(
    id_cols = c(id, issue),
    names_from = wave,
    names_prefix = "alpha_w",
    values_from = paste0(k, "_alpha")
  )

df_h15 <- df_h15 |>
  left_join(df_weights, by = c("id", "issue"))
df_h15 <- df_h15 |>
  left_join(df_p |> filter(wave==2) |> select(id, treatment_wave2), by = "id")

# _{n,q}(w2) = _{n,q}(w1) + 1 * rel_delta_var +
model_h15 <- lm(alpha_w2 ~ alpha_w1 + rel_delta_var, data = df_h15)

print(summary(model_h15))

beta1 <- coef(model_h15)["rel_delta_var"]
```

```
se_beta1 <- summary(model_h15)$coefficients["rel_delta_var", "Std. Error"]
t_stat <- summary(model_h15)$coefficients["rel_delta_var", "t value"]
p_twosided <- summary(model_h15)$coefficients["rel_delta_var", "Pr(>|t|)"]
p_onetailed <- p_twosided / 2 # one-tailed p-value for 1 < 0

cat(sprintf("=== H15 Test (beta1 < 0, %s alpha weights decreased across waves as social circle %s increased): === \n",
  sprintf(" beta1 = %.4f (+- %.4f)\n", beta1, se_beta1),
  sprintf(" t = %.4f\n", t_stat),
  sprintf(" p (one-tailed, beta1 < 0) = %.4f\n", p_onetailed),
  sprintf(" H15 supported: %s\n", beta1 < 0 && p_onetailed < 0.05))
```

Call:

```
lm(formula = alpha_w2 ~ alpha_w1 + rel_delta_var, data = df_h15)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.30054	-0.15014	-0.08524	0.08998	0.88651

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.145278	0.003744	38.798	< 2e-16 ***
alpha_w1	0.133491	0.013566	9.840	< 2e-16 ***
rel_delta_var	-0.058839	0.014342	-4.103	4.15e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2191 on 5373 degrees of freedom
(1986 observations deleted due to missingness)
Multiple R-squared: 0.01984, Adjusted R-squared: 0.01948
F-statistic: 54.38 on 2 and 5373 DF, p-value: < 2.2e-16

```
=== H15 Test (beta1 < 0, exp alpha weights decreased across waves as social circle std increased): ===
beta1 = -0.0588 (+- 0.0143)
t = -4.1025
p (one-tailed, beta1 < 0) = 0.0000
H15 supported: TRUE
```

```
# Bin rel_delta_var for cleaner visualization
df_plot <- df_h15 |>
  filter(!is.na(rel_delta_var), !is.na(alpha_w2), !is.na(alpha_w1)) |>
  mutate(
    delta_alpha = alpha_w2 - alpha_w1,
    rel_delta_var_bin = cut(rel_delta_var, breaks = 7),
    treatment_wave2 = factor(treatment_wave2, levels = c(0, 1),
                             labels = c("Control", "Treatment"))
  )

ggplot(df_plot, aes(x = rel_delta_var, y = delta_alpha, color = treatment_wave2)) +
  geom_boxplot(
    aes(group = rel_delta_var_bin),
    fill = "steelblue", alpha = 0.15,
    outlier.shape = NA,
    color = "gray40" # fixed color so boxplots don't inherit the group color
  ) +
  geom_point(alpha = 0.1, size = 1, ) +
  geom_smooth(formula = y ~ x, method = "lm", se = FALSE, alpha=0.5) +
  geom_smooth(formula = y ~ x, method = "loess", se = FALSE, linetype = "dashed", alpha=0.5) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
  scale_color_manual(
    values = c("Control" = "steelblue", "Treatment" = "firebrick"),
    name = "Wave 2"
  ) +
  labs(
```

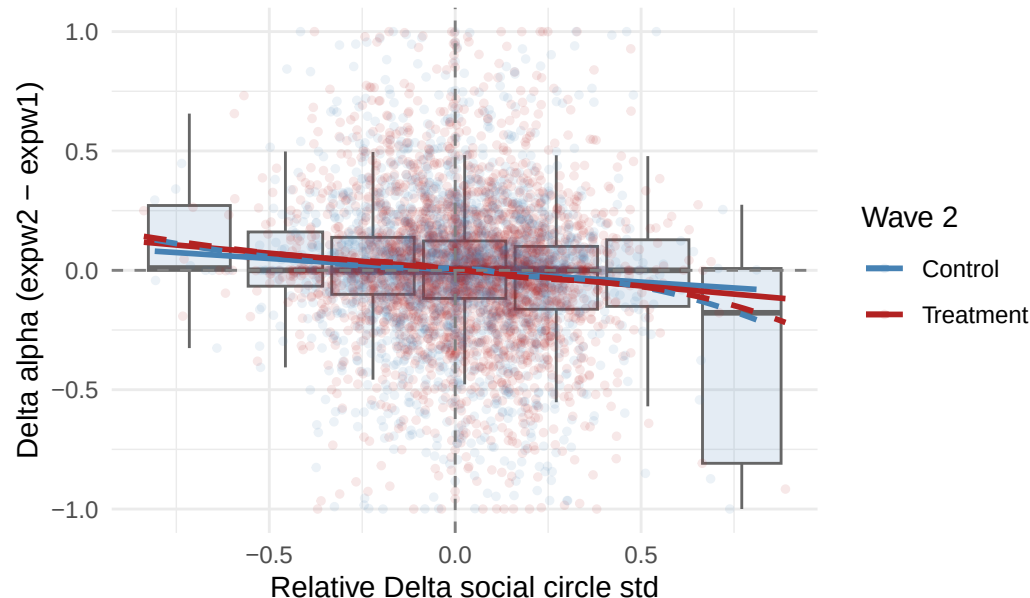
```

title = paste0("H15: Relative social-circle-", variance_type,
               " change vs. issue weight (", k, ") change"),
x      = paste0("Relative Delta social circle ", variance_type),
y      = paste0("Delta alpha (", k, "w2 - ", k, "w1)")
) +
theme_minimal()

```

Warning: Orientation is not uniquely specified when both the x and y aesthetics are continuous. Picking default orientation 'x'.

H15: Relative social-circle-std change vs. issue weight (exp)



3.4.3 H16 (caution)

3.4.3.1 H16. Increase in overall social circle variance decreases distance to typical voters.

We model the map distances between the participant themselves and typical voters of political parties r in wave $w2$ as

$$d_{n,self-r}(w2) = d_{n,self-r}(w1) + \beta_1 \cdot \mathbb{I}(\text{condition}_n = T) \cdot \sum_q \Delta x_{n,self-r,q} \cdot (\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)) + \epsilon_{n,r} \quad (20)$$

The explanatory variable is the change in social circle variance averaged over all six issues and weighted by the corresponding belief differences between participant and the typical voter of party r , $\Delta x_{n,self-r,q} = |x_{n,self,q} - x_{n,r,q}|$.

The prediction is $\beta_1 < 0$. When overall social circle variance increases, the reported distance between participant and typical voter beliefs decreases. %We test this via a t-test with significance threshold $\alpha = 0.05$.

{ERROR: This should not include the treatment. The effect we want to test is whether variance increase (weighted by opinion difference, such that) changes distances.

$$d_{n,self-r}(w2) = d_{n,self-r}(w1) + \beta_1 \cdot \sum_q \Delta x_{n,self-r,q} \cdot (\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)) + \beta_2 \cdot \text{condition}_n + u_n + \epsilon_{n,r} \quad (21)$$

or alternatively, but that's not really a fair thing to test and not really related to our theory (see discussion with Mirta):

$$d_{n,self-r}(w2) = d_{n,self-r}(w1) + \beta_1 \cdot \text{condition}_n + \epsilon_{n,r} \quad (22)$$

}

Model: Is pixel distance at all affected by treatment? ... No. clearly it's the party and individual differences

```

df_h16 <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave==2) |>
  mutate(
    group = if_else(id %in% treatment_ids, "treatment", "control"),
  ) |>
  select(pixel_dist, group, id, dot2)
model_h16 <- lmer(pixel_dist ~ group + (1|id) + (1|dot2), data = df_h16, REML=TRUE)
summary(model_h16)
anova(model_h16)

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]
Formula: pixel_dist ~ group + (1 | id) + (1 | dot2)
Data: df_h16

REML criterion at convergence: -3455.8

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.3042	-0.6556	0.0188	0.6543	3.3508

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.011489	0.1072
dot2	(Intercept)	0.001857	0.0431
Residual		0.027710	0.1665

Number of obs: 6314, groups: id, 902; dot2, 7

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	0.338016	0.017449	7.643035	19.372	8.99e-08 ***
grouptreatment	-0.006245	0.008341	900.000072	-0.749	0.454

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

(Intr)
grouptrtmnt -0.269

Type III Analysis of Variance Table with Satterthwaite's method

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
group	0.015533	0.015533	1	900	0.5606	0.4542

Model: Does previous pixel distance determine new pixel distance or does treatment ahve a direct effect?

```
df_h16_w1 <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave == 1) |>
  select(id, dot2, pixel_dist_w1 = pixel_dist)
```

```
df_h162 <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave==2) |>
  mutate(
    group = if_else(id %in% treatment_ids, "treatment", "control"),
  ) |>
  select(pixel_dist, group, id, dot2) |>
  left_join(df_h16_w1, by = c("id", "dot2"))
```

```
model_h162 <- lmer(pixel_dist ~ pixel_dist_w1 + group + (1|id) + (1|dot2), data = df_h162, REML=TRUE)
```

Warning in checkConv(attr(opt, "derivs"), opt\$par, ctrl = control\$checkConv, : Model failed to converge with max|gradi
See ?lme4::convergence and ?lme4::troubleshooting.

```
summary(model_h162)
anova(model_h162)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]
Formula: pixel_dist ~ pixel_dist_w1 + group + (1 | id) + (1 | dot2)
Data: df_h162

REML criterion at convergence: -6692.2

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.1504	-0.6018	-0.0320	0.5739	4.5649

Random effects:

Groups	Name	Variance	Std.Dev.
--------	------	----------	----------

```
id      (Intercept) 0.0088419 0.09403
dot2    (Intercept) 0.0002816 0.01678
Residual      0.0160925 0.12686
Number of obs: 6314, groups: id, 902; dot2, 7
```

Fixed effects:

```
      Estimate Std. Error      df t value Pr(>|t|)
(Intercept)  1.217e-01  8.908e-03 2.039e+01 13.665 9.93e-12 ***
pixel_dist_w1 6.108e-01  9.333e-03 6.262e+03 65.447 < 2e-16 ***
grouptreatment -2.027e-03  7.084e-03 8.958e+02 -0.286 0.775
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```
      (Intr) pxl__1
pxl_dst_w1 -0.371
grouptrtmnt -0.450 0.009
optimizer (nloptwrap) convergence code: 0 (OK)
Model failed to converge with max|grad| = 0.00280007 (tol = 0.002, component 1)
See ?lme4::convergence and ?lme4::troubleshooting.
```

Type III Analysis of Variance Table with Satterthwaite's method

```
      Sum Sq Mean Sq NumDF DenDF F value Pr(>F)
pixel_dist_w1 68.929 68.929 1 6262.2 4283.3194 <2e-16 ***
group         0.001 0.001 1 895.8 0.0819 0.7749
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

FULL MODEL!

```
variance_type = "std"

# Pull distances for wave 1 and wave 2 -----
df_h16_w1 <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave == 1) |>
  select(id, dot2, pixel_dist_w1 = pixel_dist)

df_h16_w2 <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave == 2) |>
  select(id, dot2, pixel_dist_w2 = pixel_dist)

# Pull opinion change deltaX -----
deltaX_cols <- paste0("deltaX_", questions_sc)

df_h16_dx <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave == 2) |>
  select(id, dot2, all_of(deltaX_cols))

# Pull variance change per participant per issue -----
df_h16_var <- df_p |>
  pivot_longer(
    cols = matches(paste0(variance_type, "_socialCircle_ops_")),
    names_to = c(".value", "issue"),
    names_pattern = paste0("^(", variance_type, "_socialCircle_ops)_(.*)")
  ) |>
  rename(
    var_sc = paste0(variance_type, "_socialCircle_ops")
  ) |>
  pivot_wider(
    id_cols = c(id, issue),
    names_from = wave,
    names_prefix = "w",
    values_from = var_sc
  ) |>
  rename(var_w1 = w1, var_w2 = w2) |>
  mutate(delta_var = var_w2 - var_w1)
```

```

df_h16_var_wide <- df_h16_var |>
  filter(id %in% inds_bothwaves) |>
  pivot_wider(
    id_cols      = id,
    names_from   = issue,
    names_prefix = "delta_var_",
    values_from  = delta_var
  )

# Join everything -----
df_h16 <- df_h16_dx |>
  left_join(df_h16_w1, by = c("id", "dot2")) |>
  left_join(df_h16_w2, by = c("id", "dot2")) |>
  left_join(df_h16_var_wide, by = "id") |>
  mutate(
    group = if_else(id %in% treatment_ids, "treatment", "control"),
    weighted_var_change = rowSums(
      across(all_of(deltaX_cols)) *
      across(all_of(paste0("delta_var_", questions_sc))),
      na.rm = TRUE
    ),
  )

# Fit model -----
# d_w2 = d_w1 + 1 * weighted_var_change + beta2 * treatment + random-effects +
model_h16 <- lmer(pixel_dist_w2 ~ pixel_dist_w1 + weighted_var_change + group + (1|id), data = df_h16, REML=TRUE)

print(summary(model_h16))

# -- 5. Test H16: _wvc < 0 -----
beta1      <- fixef(model_h16)["weighted_var_change"]
se_beta1   <- summary(model_h16)$coefficients["weighted_var_change", "Std. Error"]
t_stat     <- summary(model_h16)$coefficients["weighted_var_change", "t value"]
p_onetailed <- summary(model_h16)$coefficients["weighted_var_change", "Pr(>|t|)"] / 2

cat(sprintf("H16 Test (beta_weighted_var_change < 0):\n"),
  sprintf("  beta = %.4f (+- %.4f)\n", beta1, se_beta1),
  sprintf("  t = %.4f\n", t_stat),
  sprintf("  p (one-tailed, beta < 0) = %.4f\n", p_onetailed),
  sprintf("  H16 supported: %s\n", beta1 < 0 && p_onetailed < 0.05))

```

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: pixel_dist_w2 ~ pixel_dist_w1 + weighted_var_change + group +
  (1 | id)
Data: df_h16

```

REML criterion at convergence: -6613.8

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.1895	-0.5964	-0.0365	0.5728	4.4330

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.008931	0.0945
Residual		0.016325	0.1278

Number of obs: 6314, groups: id, 902

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	1.152e-01	6.250e-03	1.510e+03	18.428	<2e-16 ***
pixel_dist_w1	6.289e-01	9.184e-03	6.288e+03	68.476	<2e-16 ***
weighted_var_change	8.218e-03	4.634e-03	3.566e+03	1.773	0.0762 .

```
groupreatment      -2.897e-03  7.145e-03  8.976e+02  -0.405  0.6853
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
      (Intr) pxl__1 wght__
pixl_dst_w1 -0.519
wghtd_vr_ch  0.018 -0.063
grouptrtmnt -0.645  0.014 -0.079
H16 Test (beta_weighted_var_change < 0):
  beta = 0.0082 (+- 0.0046)
  t = 1.7734
  p (one-tailed, beta < 0) = 0.0381
H16 supported: FALSE
```

3.4.4 H17 (caution)

3.4.4.1 H17. Increase in overall social circle variance increases likeability rating of typical voters.

Similarly, we model how participant n rates the likeability rating of typical voters of party r in wave 2 as

$$y_{n,r} = \beta_0 + \beta_1 \cdot \mathbb{I}(\text{condition}_n = T) \cdot \sum_q \Delta x_{n,\text{self}-r,q} \cdot (\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)) + u_n + \epsilon_{n,r} \tag{23}$$

The prediction is $\beta_1 > 0$. When overall social circle variance increases, the likeability rating of typical voters of the political parties increases. %We test this via a t-test with significance threshold $\alpha = 0.05$.

Most simple: Does treatment affect liking? No obviously not!

```
df_h17 <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave==2) |>
  mutate(
    group = if_else(id %in% treatment_ids, "treatment", "control"),
  ) |>
  select(sympathy, group, id, dot2)
model_h17 <- lmer(sympathy ~ group + (1|id) + (1|dot2), data = df_h17, REML=TRUE)
summary(model_h17)
anova(model_h17)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: sympathy ~ group + (1 | id) + (1 | dot2)
Data: df_h17
```

REML criterion at convergence: 3196.5

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-1.84716	-0.95255	0.04288	0.71072	2.43122

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.003787	0.06154
dot2	(Intercept)	0.004716	0.06868
Residual		0.093166	0.30523

Number of obs: 6314, groups: id, 902; dot2, 7

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	0.423892	0.026778	6.507399	15.830	1.94e-06 ***
groupreatment	-0.003303	0.008775	899.999845	-0.376	0.707

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
 (Intr)
grouptrtmnt -0.184
Type III Analysis of Variance Table with Satterthwaite's method

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
group	0.013196	0.013196	1	900	0.1416	0.7067

Full Model

```
# -- 1. Pull liking for wave 1 and wave 2 -----
df_h17_w2 <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave == 2) |>
  select(id, dot2, like_w2 = sympathy)

# -- 2. Pull deltaX and variance change (identical to H16) -----
df_h17_dx <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave == 2) |>
  select(id, dot2, all_of(deltaX_cols))

# -- 3. Pull variance change (reuse from H16) -----
# df_h16_var_wide already computed above - reuse directly

# -- 4. Join and compute weighted variance change -----
df_h17 <- df_h17_dx |>
  left_join(df_h17_w2, by = c("id", "dot2")) |>
  left_join(df_h16_var_wide, by = "id") |>
  mutate(
    group = if_else(id %in% treatment_ids, "treatment", "control"),
    weighted_var_change = rowSums(
      across(all_of(deltaX_cols)) *
      across(all_of(paste0("delta_var_", questions_sc))),
      na.rm = TRUE
    )
  )

# -- 6. Fit model -----
# like_w2 = 0 + _wvc * weighted_var_change + _group * group + (1|id) +
model_h17 <- lmer(like_w2 ~ weighted_var_change + group + (1|id),
  data = df_h17, REML = TRUE)
print(summary(model_h17))

# -- 7. Test H17: _wvc > 0 -----
beta1 <- fixef(model_h17)["weighted_var_change"]
se_beta1 <- summary(model_h17)$coefficients["weighted_var_change", "Std. Error"]
t_stat <- summary(model_h17)$coefficients["weighted_var_change", "t value"]
p_onetailed <- summary(model_h17)$coefficients["weighted_var_change", "Pr(>|t|)"] / 2

cat(sprintf("H17 Test (beta_wvc > 0):\n"),
  sprintf("  beta = %.4f (+- %.4f)\n", beta1, se_beta1),
  sprintf("  t = %.4f\n", t_stat),
  sprintf("  p (one-tailed, beta > 0) = %.4f\n", p_onetailed),
  sprintf("  H17 supported: %s\n", beta1 > 0 && p_onetailed < 0.05))
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]
Formula: like_w2 ~ weighted_var_change + group + (1 | id)
Data: df_h17

REML criterion at convergence: 3432.6

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-1.64411	-1.02121	0.09116	0.72275	2.07669

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.003522	0.05934
Residual		0.097305	0.31194

Number of obs: 6314, groups: id, 902

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	4.245e-01	6.643e-03	8.748e+02	63.903	< 2e-16 ***
weighted_var_change	-3.021e-02	7.490e-03	1.508e+03	-4.033	5.79e-05 ***
grouptreatment	3.537e-04	8.905e-03	8.790e+02	0.040	0.968

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	wght__
wghtd_vr_ch	-0.023	
grouptrtmnt	-0.743	-0.102

H17 Test (beta_wvc > 0):

beta = -0.0302 (+- 0.0075)

t = -4.0329

p (one-tailed, beta > 0) = 0.0000

H17 supported: FALSE

3.5 NEW: new versions of H16 and H17 (moderation)

3.5.1 H16 new

3.5.1.1 H16: Treatment increases social circle variance, which weakens the effect of opinion differences on perceived distance.

We test this hypothesis in two steps, conditional on H14 being confirmed (i.e., the nudge successfully increased social circle variance in the treatment group).

Step 1 — Intent-to-Treat (ITT): Does the treatment moderate the relationship between opinion differences $\Delta X_{n,r,q} = |x_{n,\text{self},q} - x_{n,r,q}|$ and perceived distance d ? We model:

$$d_{n,\text{self}-r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \mathbb{I}(\text{condition}_n = T) + \beta_3 \cdot \Delta X_{n,r,q} \cdot \mathbb{I}(\text{condition}_n = T) + u_n + v_r + v_q + \epsilon_{n,r,q} \tag{24}$$

where u_n is the participant random intercept, v_r the party voter random intercept, and v_q the question random intercept. The prediction is $\beta_3 < 0$: the treatment weakens the effect of opinion differences on perceived distance.

Step 2 — Variance as moderator: Does the actual change in variance $\Delta \text{var}_{n,q} = \text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)$ moderate the $\Delta X \rightarrow d$ relationship, above and beyond treatment assignment?

$$d_{n,\text{self}-r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \Delta \text{var}_{n,q} + \beta_3 \cdot \Delta X_{n,r,q} \cdot \Delta \text{var}_{n,q} + u_n + v_r + v_q + \epsilon_{n,r,q} \tag{25}$$

The prediction is $\beta_3 < 0$: when social circle variance increases, the effect of opinion differences on perceived distance weakens. Step 2 speaks to the *mechanism*: it is the variance increase itself, not merely treatment assignment, that drives the moderation. Together, Steps 1 and 2 establish the causal chain:

$$\mathbb{I}(T) \xrightarrow{H14} \Delta \text{var} \xrightarrow{\beta_3 < 0} \text{weakens } (\Delta X \rightarrow d) \tag{26}$$

```
variance_type <- "std"
var_col       <- paste0(variance_type, "_socialCircle_ops")
q_pattern     <- paste(questions_sc, collapse = "|")

# -- 1. Build long dataset (wave 2 only) for Step 1 (ITT) -----
df_h16_itt <- df_diff |>
  filter(
    dot1 == "self",
    dot2 %in% partiesVars,
    wave == 2,
    id %in% c(treatment_ids, control_ids)
  ) |>
  mutate(
    group = case_when(
      id %in% treatment_ids ~ "treatment",
      id %in% control_ids   ~ "control"
    ),
  ) |>
  pivot_longer(
    cols = matches(
      paste0("(^deltaX|", var_col, ")_(", q_pattern, ")$")
    ),
```

```

    names_to      = c(".value", "issue"),
    names_pattern = paste0("^(deltaX|", var_col, ")_(", q_pattern, ")$")
  ) |>
  select(id, wave, dot2, group, pixel_dist, issue, deltaX, all_of(var_col)) |>
  mutate(treatment = as.numeric(id %in% treatment_ids))

# -- 2. Build delta_var from df_p (waves 1 & 2) for Step 2 -----
df_delta_var <- df_p |>
  filter(id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(
    cols      = matches(paste0(var_col, "_")),
    names_to   = c(".value", "issue"),
    names_pattern = paste0("^((", var_col, ")_(.*)")
  ) |>
  pivot_wider(
    id_cols      = c(id, issue),
    names_from    = wave,
    names_prefix  = "var_w",
    values_from   = all_of(var_col)
  ) |>
  mutate(
    delta_var     = var_w2 - var_w1,
    delta_var_centered = as.numeric(scale(delta_var, center = TRUE, scale = FALSE))
  ) |>
  select(id, issue, delta_var, delta_var_centered)

# -- 3. Join delta_var into wave-2 dataset for Step 2 -----
df_h16_var <- df_h16_itt |>
  left_join(df_delta_var, by = c("id", "issue"))

# -- Step 1: ITT -----
# Does treatment moderate the deltaX -> pixel_dist relationship?
model_h16_itt <- lmer(
  pixel_dist ~ deltaX * treatment +
    (1 | id) +
    (1 | issue) +
    (1 | dot2),
  data = df_h16_itt
)
summary(model_h16_itt)

cat(sprintf(
  "\nITT interaction (beta_3): %.4f\nPrediction (beta_3 < 0): %s\n",
  fixef(model_h16_itt)["deltaX:treatment"],
  fixef(model_h16_itt)["deltaX:treatment"] < 0
))

# -- Step 2: Variance as moderator -----
# Does actual delta_var moderate the deltaX -> pixel_dist relationship?
model_h16_var <- lmer(
  pixel_dist ~ deltaX * delta_var_centered +
    (1 | id) +
    (1 | issue) +
    (1 | dot2),
  data = df_h16_var
)
summary(model_h16_var)

cat(sprintf(
  "\nVariance moderation (beta_3): %.4f\nPrediction (beta_3 < 0): %s\n",

```

```
fixef(model_h16_var)["deltaX:delta_var_centered"],
fixef(model_h16_var)["deltaX:delta_var_centered"] < 0
))
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: pixel_dist ~ deltaX * treatment + (1 | id) + (1 | issue) + (1 |
dot2)
Data: df_h16_itt

REML criterion at convergence: -33909.2

Scaled residuals:
Min 1Q Median 3Q Max
-4.5669 -0.6155 0.0321 0.6615 3.8564

Random effects:
Groups Name Variance Std.Dev.
id (Intercept) 1.441e-02 0.120024
dot2 (Intercept) 1.784e-03 0.042233
issue (Intercept) 4.494e-05 0.006704
Residual 2.204e-02 0.148460
Number of obs: 37884, groups: id, 902; dot2, 7; issue, 6

Fixed effects:
Estimate Std. Error df t value Pr(>|t|)
(Intercept) 2.810e-01 1.738e-02 8.413e+00 16.164 1.25e-07 ***
deltaX 9.067e-02 2.279e-03 3.739e+04 39.784 < 2e-16 ***
treatment -1.068e-02 8.420e-03 9.992e+02 -1.268 0.205
deltaX:treatment 4.359e-03 2.996e-03 3.741e+04 1.455 0.146

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
(Intr) deltaX trtmnt
deltaX -0.083
treatment -0.272 0.168
dltX:trtmnt 0.062 -0.751 -0.227

ITT interaction (beta_3): 0.0044
Prediction (beta_3 < 0): FALSE
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: pixel_dist ~ deltaX * delta_var_centered + (1 | id) + (1 | issue) +
(1 | dot2)
Data: df_h16_var

REML criterion at convergence: -33908.7

Scaled residuals:
Min 1Q Median 3Q Max
-4.5631 -0.6153 0.0319 0.6605 3.8425

Random effects:
Groups Name Variance Std.Dev.
id (Intercept) 1.439e-02 0.119965
dot2 (Intercept) 1.787e-03 0.042274
issue (Intercept) 4.486e-05 0.006698
Residual 2.204e-02 0.148464
Number of obs: 37884, groups: id, 902; dot2, 7; issue, 6

Fixed effects:
Estimate Std. Error df t value Pr(>|t|)
(Intercept) 2.749e-01 1.674e-02 7.189e+00 16.423 5.8e-07 ***
deltaX 9.317e-02 1.505e-03 3.703e+04 61.904 < 2e-16 ***

```
delta_var_centered      -8.828e-03  7.265e-03  3.767e+04 -1.215    0.224
deltaX:delta_var_centered 3.455e-03  7.066e-03  3.722e+04  0.489    0.625
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
      (Intr) deltaX dlt_v_
deltaX      -0.057
dlt_vr_cntr  0.001  0.003
dltX:dlt_v_  0.000 -0.023 -0.704

Variance moderation (beta_3): 0.0035
Prediction (beta_3 < 0): FALSE
```

3.5.2 H17 new

3.5.2.1 H17: Treatment increases social circle variance, which increases likeability ratings of typical party voters.

We test this hypothesis in two steps, conditional on H14 being confirmed (i.e., the nudge successfully increased social circle variance in the treatment group).

Step 1 — Intent-to-Treat (ITT): Does the treatment moderate the relationship between opinion differences $\Delta X_{n,r,q}$ and likeability ratings $y_{n,r}$? We model:

$$y_{n,r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \mathbb{I}(\text{condition}_n = T) + \beta_3 \cdot \Delta X_{n,r,q} \cdot \mathbb{I}(\text{condition}_n = T) + u_n + v_r + v_q + \epsilon_{n,r,q} \tag{27}$$

where u_n is the participant random intercept, v_r the party voter random intercept, and v_q the question random intercept. The prediction is $\beta_3 > 0$: the treatment weakens the negative effect of opinion differences on likeability, i.e. participants in the treatment condition rate typical voters of parties they disagree with more favourably.

Step 2 — Variance as moderator: Does the actual change in variance $\Delta \text{var}_{n,q} = \text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)$ moderate the $\Delta X \rightarrow y$ relationship, above and beyond treatment assignment?

$$y_{n,r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \Delta \text{var}_{n,q} + \beta_3 \cdot \Delta X_{n,r,q} \cdot \Delta \text{var}_{n,q} + u_n + v_r + v_q + \epsilon_{n,r,q} \tag{28}$$

The prediction is $\beta_3 > 0$: when social circle variance increases, the negative effect of opinion differences on likeability weakens. Step 2 speaks to the *mechanism*: it is the variance increase itself, not merely treatment assignment, that drives the moderation. Together, Steps 1 and 2 establish the causal chain:

$$\mathbb{I}(T) \xrightarrow{H14} \Delta \text{var} \xrightarrow{\beta_3 > 0} \text{weakens } (\Delta X \rightarrow -y) \tag{29}$$

```
variance_type <- "std"
var_col       <- paste0(variance_type, "_socialCircle_ops")
q_pattern     <- paste(questions_sc, collapse = "|")

# -- 1. Build long dataset (wave 2 only) for Step 1 (ITT) -----
# Note: replace `likeability` with the actual column name in your data
df_h17_itt <- df_diff |>
  filter(
    dot1 == "self",
    dot2 %in% partiesVars,
    wave == "2",
    id %in% c(treatment_ids, control_ids)
  ) |>
  pivot_longer(
    cols = matches(
      paste0("^(\deltaX|", var_col, ")_(", q_pattern, ")$")
    ),
    names_to = c(".value", "issue"),
    names_pattern = paste0("^(\deltaX|", var_col, ")_(", q_pattern, ")$")
  ) |>
  select(id, wave, dot2, sympathy, issue, deltaX, all_of(var_col)) |>
  mutate(treatment = as.numeric(id %in% treatment_ids))

# -- 2. Reuse delta_var from H16 pipeline -----
# If running independently, rebuild it here:
df_delta_var <- df_p |>
  filter(id %in% c(treatment_ids, control_ids)) |>
```

```

pivot_longer(
  cols          = matches(paste0(var_col, "_")),
  names_to      = c(".value", "issue"),
  names_pattern = paste0("^(", var_col, ")_(.*)")
) |>
pivot_wider(
  id_cols      = c(id, issue),
  names_from   = wave,
  names_prefix = "var_w",
  values_from  = all_of(var_col)
) |>
mutate(
  delta_var      = var_w2 - var_w1,
  delta_var_centered = as.numeric(scale(delta_var, center = TRUE, scale = FALSE))
) |>
select(id, issue, delta_var, delta_var_centered)

```

```

# -- 3. Join delta_var into wave-2 dataset for Step 2 -----
df_h17_var <- df_h17_itt |>
  left_join(df_delta_var, by = c("id", "issue"))

```

```

# -- Step 1: ITT -----
# Does treatment moderate the deltaX -> likeability relationship?
model_h17_itt <- lmer(
  sympathy ~ deltaX * treatment +
    (1 | id) +
    (1 | issue) +
    (1 | dot2),
  data = df_h17_itt
)
summary(model_h17_itt)

cat(sprintf(
  "\nITT interaction (beta_3): %.4f\nPrediction (beta_3 > 0): %s\n",
  fixef(model_h17_itt)["deltaX:treatment"],
  fixef(model_h17_itt)["deltaX:treatment"] > 0
))

```

```

# -- Step 2: Variance as moderator -----
# Does actual delta_var moderate the deltaX -> likeability relationship?
model_h17_var <- lmer(
  sympathy ~ deltaX * delta_var_centered +
    (1 | id) +
    (1 | issue) +
    (1 | dot2),
  data = df_h17_var
)
summary(model_h17_var)

cat(sprintf(
  "\nVariance moderation (beta_3): %.4f\nPrediction (beta_3 > 0): %s\n",
  fixef(model_h17_var)["deltaX:delta_var_centered"],
  fixef(model_h17_var)["deltaX:delta_var_centered"] > 0
))

```

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: sympathy ~ deltaX * treatment + (1 | id) + (1 | issue) + (1 |
dot2)
Data: df_h17_itt

```

REML criterion at convergence: 8981.1

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-3.1623	-0.7003	-0.0553	0.6333	4.3067

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.0136288	0.11674
dot2	(Intercept)	0.0038240	0.06184
issue	(Intercept)	0.0002323	0.01524
Residual		0.0702531	0.26505

Number of obs: 37884, groups: id, 902; dot2, 7; issue, 6

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	5.494e-01	2.510e-02	7.880e+00	21.884	2.44e-08 ***
deltaX	-1.994e-01	4.037e-03	3.786e+04	-49.391	< 2e-16 ***
treatment	1.111e-02	8.964e-03	1.214e+03	1.240	0.21525
deltaX:treatment	-1.680e-02	5.311e-03	3.786e+04	-3.163	0.00156 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr) deltaX	trtmnt
deltaX	-0.101	
treatment	-0.200	0.280
dltX:trtmnt	0.076	-0.751

ITT interaction (beta_3): -0.0168

Prediction (beta_3 > 0): FALSE

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]

Formula: sympathy ~ deltaX * delta_var_centered + (1 | id) + (1 | issue) + (1 | dot2)

Data: df_h17_var

REML criterion at convergence: 8981.6

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-3.2042	-0.6994	-0.0543	0.6339	4.3527

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.0135473	0.11639
dot2	(Intercept)	0.0037943	0.06160
issue	(Intercept)	0.0002318	0.01522
Residual		0.0702667	0.26508

Number of obs: 37884, groups: id, 902; dot2, 7; issue, 6

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	5.557e-01	2.450e-02	7.351e+00	22.678	4.5e-08 ***
deltaX	-2.090e-01	2.669e-03	3.773e+04	-78.299	< 2e-16 ***
delta_var_centered	3.553e-02	1.280e-02	3.756e+04	2.775	0.00552 **
deltaX:delta_var_centered	-2.561e-02	1.257e-02	3.761e+04	-2.038	0.04153 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr) deltaX	dlt_v_
deltaX	-0.070	
dlt_vr_cntr	0.001	0.002
dltX:dlt_v_	0.000	-0.023

Variance moderation (beta_3): -0.0256
Prediction (beta_3 > 0): FALSE
#

4 Exclusion criteria

Participants are excluded if they complete the survey in less than one third of the median completion time, provide invariant slider responses, or do not indicate their own beliefs on the six political issues.